A Thermodynamic Obstruction to Finite-Time Blow-Up in the 3D Navier-Stokes Equations (Conditional Framework)

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Abstract

We propose a doubly conditional regularity framework for the 3D incompressible Navier-Stokes equations based on a thermodynamic distance functional. Rather than bounding the velocity gradient globally, we construct an attractor-distance functional $H(u \mid \mathcal{A})$ measuring the L^2-distance of the active flow from the global Leray–Hopf attractor. Under two conditions—(**Assumption G**) that the time-dependent minimising projection onto the attractor is sufficiently regular, and (**Condition D**) that the cumulative viscous dissipation dominates the Gronwall cost near a hypothetical blow-up time—we show that finite-time blow-up would force H below zero, contradicting its non-negativity. **Importantly, Condition D at the L^2-level is a structural limitation:** the Leray energy identity bounds the dissipation integral from above, while the Gronwall cost grows exponentially, making Condition D increasingly difficult (and plausibly impossible) to satisfy for slow blow-up. The argument therefore identifies the H^1-level as the natural Sobolev level for the obstruction strategy, where the enstrophy dissipation has no a priori upper bound. We identify sufficient geometric conditions for G (positive reach, log-Lipschitz projections, BV selections) and present an H^1-level lift where the enstrophy dissipation can serve as the divergent term; the H^1-programme introduces additional open problems (Bihari-type nonlinear Gronwall, enstrophy-dissipation divergence) that are discussed in detail. This paper establishes a *doubly conditional framework*: under the joint validity of Assumption G (BV-regular selection on the attractor) and Condition D (dissipation dominance near blow-up), finite-time blow-up is excluded (Theorem 3.35). This is *not* a claim of a new unconditional regularity result; the value lies in the decomposition of the regularity question into two structurally distinct components—one geometric (Assumption G), one analytical (Condition D)—and in identifying concrete attack vectors for each. The companion paper on log-distance integrability [41] provides sufficient geometric conditions for Assumption G.

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1 Introduction

The classical challenge concerning the 3D incompressible Navier-Stokes equations is to preclude the formation of finite-time singularities (blow-ups) in the velocity field $u(x,t) \in \mathbb{R}^3$, a problem that has remained open since Leray's foundational work [7]. The flow is governed by:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad \text{on } \Omega \subset \mathbb{R}^3. \tag{1}$$

The well-known Beale–Kato–Majda (BKM) criterion [1] states that a singularity at time T^* occurs if and only if the vorticity diverges:

$$\int_0^{T^*} \|\nabla \times u(t)\|_{L^\infty} \, dt = \infty. \tag{2}$$

In this paper, we construct a thermodynamic Lyapunov functional measuring the distance of the active flow from the global Leray–Hopf attractor. We show that, *conditional on a regularity assumption about the attractor projection* (Assumption G, Section 3) and a *dissipation-dominance condition* (D), any hypothetical blow-up leads to a contradiction: the enstrophy diverges pointwise, driving the non-negative distance functional below zero whenever the cumulative viscous dissipation outpaces the Gronwall cost. The resulting doubly conditional framework decomposes the regularity question into two structurally distinct open problems: the geometry of the Navier–Stokes attractor (Assumption G) and the relative blow-up rates of dissipation versus Gronwall factors (condition D). We do not claim that either condition is easier to verify than regularity itself; the value lies in the decomposition and in identifying concrete attack vectors for each component.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

2 The Energy-Dissipation Divergence Functional

We define a generalized divergence functional $H(u \mid \mathcal{A})$ representing the distance of the active velocity field u from the global attractor $\mathcal{A}$.

Definition 2.1 (Divergence from the Global Attractor)**.** Let $\Omega = \mathbb{T}^3$ be the flat torus. We define the reference manifold $\mathcal{A}$ as the global Leray–Hopf attractor of the Navier–Stokes semiflow (cf. [5, 8]). For any active flow field $u \in L^2(\Omega)$, the distance functional is:

$$H(u \mid \mathcal{A}) := \inf_{v \in \mathcal{A}} E_v(u) = \inf_{v \in \mathcal{A}} \frac{1}{2} \int_\Omega |u - v|^2 \, dx. \tag{3}$$

Because $\mathcal{A}$ is a compact subset of $L^2(\Omega)$, the infimum is achieved. We denote by $v(t) \in \mathcal{A}$ a measurable selection that minimizes this distance at time t, such that $H(u \mid \mathcal{A}) = \frac{1}{2} \int_\Omega |w|^2 \, dx$ with $w := u - v$.

Proposition 2.2 (Regularity of the global attractor)**.** *Let $\Omega = \mathbb{T}^3$ and let $f \in L_t^\infty H_x^1(\Omega)$ be smooth forcing. The global Leray–Hopf attractor $\mathcal{A}$ satisfies:*

(i) *$\mathcal{A}$ is compact in $H^1(\Omega)$.*

(ii) *Every $v \in \mathcal{A}$ is smooth: $v \in C^\infty(\Omega)$.*

(iii) $\sup_{v\in\mathcal{A}} \|v\|_{W^{1,\infty}} < \infty$.

Proof. **(i) Compactness in H^1.** The Navier–Stokes semiflow $S(t)$ on $\mathbb{T}^3$ with forcing $f \in L^\infty_t H^1_x$ possesses an absorbing ball $\mathcal{B}_1 \subset H^1(\Omega)$: there exists $R_1 > 0$ (depending on ν and $\|f\|_{L^\infty H^1}$) and $t_1 > 0$ such that $S(t)\big(\mathcal{B}_0\big) \subset \mathcal{B}_1 := \{u : \|u\|_{H^1} \le R_1\}$ for all $t \ge t_1$ and every bounded set $\mathcal{B}_0 \subset L^2(\Omega)$. This follows from the standard energy estimate: testing the Navier–Stokes equation with $-\Delta u$ yields a differential inequality for $\|\nabla u\|^2_{L^2}$ that, combined with the Poincaré inequality, produces uniform-in-time H^1 bounds after a transient (cf. [8], Chapter III, §3; [5], §9.2).

The global attractor is defined as $\mathcal{A} = \bigcap_{t\ge0} S(t)\big(\mathcal{B}_1\big)$, and it is the maximal compact invariant set of the semiflow. On $\mathbb{T}^3$, the semiflow $S(t)$ is *compact* as a map from H^1 to H^1 (indeed, it maps bounded sets in L^2 into bounded sets in H^2 for $t > 0$, and the embedding $H^2(\mathbb{T}^3) \hookrightarrow H^1(\mathbb{T}^3)$ is compact by the Rellich–Kondrachov theorem). By the abstract attractor existence theorem ([8], Theorem I.1.1; [5], Theorem 9.6), $\mathcal{A}$ is compact in $H^1(\Omega)$.

(ii) Smoothness of attractor elements. Let $v \in \mathcal{A}$. Since $\mathcal{A}$ is invariant, there exists a complete bounded trajectory $\{v(t)\}_{t\in\mathbb{R}} \subset \mathcal{A}$ passing through $v = v(0)$. Each $v(t)$ satisfies the Navier–Stokes equation

$$\partial_t v + (v \cdot \nabla)v + \nabla p = \nu\Delta v + f.$$

Since $v(t) \in H^1(\mathbb{T}^3)$ uniformly and $f \in H^1$, we bootstrap regularity as follows.

First step ($H^1 \to H^2$): The nonlinear term $(v\cdot\nabla)v \in L^2(\mathbb{T}^3)$ by the Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$ and Hölder's inequality. Rewriting the equation as the stationary Stokes system $-\nu\Delta v + \nabla p = f - (v \cdot \nabla)v - \partial_t v$, elliptic regularity for the Stokes operator on $\mathbb{T}^3$ ([4], §2.4; [8], §II.3) yields $v(t) \in H^2(\mathbb{T}^3)$.

Inductive step ($H^k \to H^{k+1}$): Suppose $v \in H^k$ for some $k \ge 2$. Since $H^k(\mathbb{T}^3)$ is a Banach algebra for $k \ge 2$ (by the Sobolev embedding $H^k \hookrightarrow L^\infty$ for $k > 3/2$), the nonlinear term $(v \cdot \nabla)v \in H^{k-1}$. Elliptic regularity for the Stokes operator then gives $v \in H^{k+1}$. By induction, $v \in \bigcap_{k\ge1} H^k(\mathbb{T}^3) = C^\infty(\mathbb{T}^3)$.

(iii) Uniform $W^{1,\infty}$ bound. By part (ii) and the bootstrap argument, $\mathcal{A} \subset H^k(\mathbb{T}^3)$ for every $k \ge 1$. Moreover, the absorbing ball argument extends to higher Sobolev norms: for each k, there exists $R_k > 0$ such that $\sup_{v\in\mathcal{A}} \|v\|_{H^k} \le R_k$ (cf. [5], Theorem 12.1; [8], §III.5). Choosing $k > 5/2$ (e.g. $k = 3$), the Sobolev embedding theorem on $\mathbb{T}^3$ gives $H^k(\mathbb{T}^3) \hookrightarrow W^{1,\infty}(\mathbb{T}^3)$ continuously. Therefore

$$\sup_{v\in\mathcal{A}} \|v\|_{W^{1,\infty}} \;\le\; C_{\mathrm{Sob}} \sup_{v\in\mathcal{A}} \|v\|_{H^k} \;\le\; C_{\mathrm{Sob}}\, R_k \;<\; \infty,$$

where C_{Sob} is the embedding constant. □

Remark 2.3 (On the attractor in the absence of unconditional regularity). The existence of a compact global attractor for the 3D Navier–Stokes semiflow on $\mathbb{T}^3$ is standard when the semiflow is known to generate a continuous dynamical system on H^1 (cf. [8, 5]). In the absence of an unconditional regularity proof, the semiflow is only defined through Leray–Hopf weak solutions, for which uniqueness remains open. Nevertheless, the *weak attractor* (the ω-limit set of the absorbing ball under the Leray–Hopf semiflow; see [5], §12) exists and is compact in L^2, and every element on the weak attractor is smooth by the standard bootstrap argument (invariance gives a complete bounded trajectory, which regularises via the Stokes operator). The present paper works entirely within this framework: the attractor $\mathcal{A}$ is the weak Leray–Hopf attractor, and the smoothness of its elements (Proposition 2.2(ii)–(iii)) does not presuppose global regularity of arbitrary solutions.

Selection dependence. Since uniqueness of Leray–Hopf solutions is open, the semiflow $S(t)$ depends on the choice of solution selection, and different selections may produce different attractors. The present framework applies to any fixed selection that generates a semiflow whose ω-limit set is compact in H^1 (which is the case for any strong selection, but remains open for pathological weak selections). The conclusions of Theorem 3.35 hold for every such attractor, provided the corresponding nearest-point projection satisfies Assumption G. This is consistent with the Sell–Chepyzhov–Vishik trajectory attractor approach [37], which avoids the uniqueness issue entirely by working with sets of complete trajectories rather than a single-valued semiflow.

By tracking the evolution of this functional, we establish an exact dissipation bound.

Lemma 2.4 (Energy-Dissipation Divergence Bound)**.** *Let $u \in L_t^\infty H_x^1(\Omega)$ be a divergence-free Leray–Hopf solution to the Navier-Stokes equations with smooth forcing f, and let $v \in L_t^\infty H_x^1(\Omega)$ be a divergence-free field (not necessarily an NS solution) with $\partial_t v$ locally integrable in time. Define the residual $\mathcal{R}_v := \partial_t v + (v\cdot\nabla)v - \nu\Delta v - f$, which vanishes when v is itself an NS solution. The time evolution of $E_v(t) = \frac{1}{2}\|u-v\|_{L^2}^2$ is bounded by:*

$$\frac{d}{dt}E_v(t) \leq -\frac{\nu}{2}\|\nabla u\|_{L^2}^2 + C_1(t)E_v(t) + C_2(t), \tag{4}$$

where $C_1(t) = 2\|\nabla v\|_{L^\infty} + 1$ and $C_2(t) = \frac{1}{2}\|(v\cdot\nabla)v\|_{L^2}^2 + \frac{1}{2}\|f - \partial_t v + \mathcal{R}_v\|_{L^2}^2 + \frac{\nu}{2}\|\nabla v\|_{L^2}^2$. When v is an NS solution, $\mathcal{R}_v = 0$ and C_2 simplifies to $\frac{1}{2}\|(v\cdot\nabla)v\|_{L^2}^2 + \frac{1}{2}\|f-\partial_t v\|_{L^2}^2 + \frac{\nu}{2}\|\nabla v\|_{L^2}^2$. When $v(t)$ is the time-varying attractor minimiser (Remark 2.5), the term $\partial_t v$ must be interpreted as the total rate of change of the selection, and $\mathcal{R}_v \neq 0$ in general; the BV modification in the proof of Theorem 3.35 (equation (28)) handles this case.

Proof. We subtract the governing equations for v from u, yielding the evolution equation for $w = u - v$:

$$\frac{d}{dt}E_v(t) = \int_\Omega w\cdot\partial_t w\,dx = \int_\Omega w\cdot(-(u\cdot\nabla)u - \nabla p + \nu\Delta u + f - \partial_t v)\,dx. \tag{5}$$

Because $\nabla\cdot w = 0$ and Ω is periodic, the pressure term vanishes: $\int_\Omega w\cdot\nabla p\,dx = 0$. Integration by parts on the viscous term gives:

$$\nu\int_\Omega w\cdot\Delta u\,dx = -\nu\int_\Omega \nabla w : \nabla u\,dx = -\nu\|\nabla u\|_{L^2}^2 + \nu\int_\Omega \nabla v : \nabla u\,dx. \tag{6}$$

Applying the Cauchy-Schwarz and Young inequalities yields:

$$\nu\int_\Omega w\cdot\Delta u\,dx \leq -\frac{\nu}{2}\|\nabla u\|_{L^2}^2 + \frac{\nu}{2}\|\nabla v\|_{L^2}^2. \tag{7}$$

For the critical convective integral $I := \int_\Omega w\cdot(u\cdot\nabla)u\,dx$, we decompose $u = v+w$. Utilizing the fundamental cancellation property $\int_\Omega w\cdot(y\cdot\nabla)w\,dx = 0$ for any divergence-free y, we isolate the remaining non-zero terms:

$$-I = -\int_\Omega w\cdot(v\cdot\nabla)v\,dx - \int_\Omega w\cdot(w\cdot\nabla)v\,dx. \tag{8}$$

These terms are bounded as:

$$|-I| \leq \|w\|_{L^2}\|(v\cdot\nabla)v\|_{L^2} + \|\nabla v\|_{L^\infty}\|w\|_{L^2}^2. \tag{9}$$

Substituting these bounds back into the time derivative equation and applying Young's inequality to the residual linear terms $\|w\|_{L^2}\|f-\partial_t v\|_{L^2}$ and $\|w\|_{L^2}\|(v\cdot\nabla)v\|_{L^2}$ yields the stated result. □

Because $\mathcal{A}$ is the global Leray-Hopf attractor on $\mathbb{T}^3$, absorbing ball estimates ensure that the uniform bounds $\sup_{v\in\mathcal{A}} C_1(t) < \infty$ and $\sup_{v\in\mathcal{A}} C_2(t) < \infty$ hold globally for all t. Taking the infimum over $\mathcal{A}$ transitions the individual energy bound into an absolute stability limit for the flow divergence $H(u \mid \mathcal{A})$.

Remark 2.5 (The minimiser is not a single NS trajectory). In the proof of Theorem 3.35, the minimising selection $v(t) \in \mathcal{A}$ is the nearest attractor element at each time t, *not* a single NS solution trajectory. As $u(t)$ evolves, the minimiser may jump between different points of $\mathcal{A}$. Lemma 2.4 is stated for a *fixed* NS solution v; when applied to the time-varying minimiser, the term $\partial_t v$ in C_2 must be interpreted as the total rate of change of the selection (including jumps), not merely the NS evolution at $v(t)$. This is precisely why Assumption G (controlling $\partial_t v$) or the BV modification (Stieltjes integral, equation (28)) is needed.

3 Main Result: Conditional Exclusion of Finite-Time Singularities

We now establish the main regularity result. Two auxiliary lemmas are stated and proved first to keep the principal argument self-contained.

Lemma A (Biot–Savart Logarithmic Estimate)

Lemma 3.1 (Biot–Savart estimate on $\mathbb{T}^3$)**.** *Let u be a periodic, divergence-free vector field on $\mathbb{T}^3$, and let $\omega = \nabla\times u$ be its vorticity. For any $s > 3/2$ there exists a constant $C_{\mathrm{BS}} = C_{\mathrm{BS}}(s,\Omega) > 0$ (depending only on s and the geometry of $\Omega = \mathbb{T}^3$, but* not *on u) such that*

$$\|\nabla u\|_{L^\infty} \;\leq\; C_{\mathrm{BS}}\,\|\omega\|_{L^\infty}\left(1+\log\frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right), \qquad \textit{whenever } \|\omega\|_{L^\infty} > 0. \tag{10}$$

Proof. We recover ∇u from ω via the periodic Biot–Savart law: $u = K * \omega$, where K is the matrix-valued convolution kernel on $\mathbb{T}^3$ arising from inverting the curl–divergence system. In particular $\nabla u = \nabla K * \omega$, and ∇K is a Calderón–Zygmund singular integral kernel (cf. [2], Chapter 2).

Step 1 (Littlewood–Paley decomposition). Let $\{\Delta_j\}_{j\geq 0}$ denote the standard Littlewood–Paley projection operators on $\mathbb{T}^3$, associated with a smooth dyadic partition of unity $\{\varphi_j\}$ in frequency space such that $\operatorname{supp}\hat{\varphi}_j \subset \{|\xi| \sim 2^j\}$ and $\sum_{j\geq 0}\varphi_j(\xi) = 1$ for all $\xi \neq 0$. We split ∇u at a free integer parameter $N \geq 1$:

$$\nabla u \;=\; \underbrace{\sum_{j=0}^{N}\Delta_j(\nabla u)}_{=:\,S_{\leq N}} \;+\; \underbrace{\sum_{j=N+1}^{\infty}\Delta_j(\nabla u)}_{=:\,S_{>N}}.$$

Step 2 (Low frequencies $j \le N$). Because ∇K is a Calderón–Zygmund operator of order zero on $\mathbb{T}^3$, it maps L^∞ to BMO, and on each Littlewood–Paley block Δ_j the Bernstein embedding BMO $\hookrightarrow L^\infty$ (at fixed frequency scale) yields

$$\|\Delta_j(\nabla u)\|_{L^\infty} \;\le\; C_{\mathrm{CZ}}\,\|\Delta_j\omega\|_{L^\infty} \;\le\; C_{\mathrm{CZ}}\,\|\omega\|_{L^\infty},$$

where C_{CZ} depends only on the operator norm of ∇K and the geometry of $\mathbb{T}^3$. Summing over $j = 0, \dots, N$:

$$\|S_{\le N}\|_{L^\infty} \;\le\; C_{\mathrm{CZ}}\,(N+1)\,\|\omega\|_{L^\infty}. \tag{11}$$

Step 3 (High frequencies $j > N$). For each dyadic block with Fourier support in $\{|\xi| \sim 2^j\}$, the Bernstein inequality on $\mathbb{T}^3$ gives (with $p = 2$, $q = \infty$):

$$\|\Delta_j(\nabla u)\|_{L^\infty} \;\le\; C_B\,2^{3j/2}\,\|\Delta_j(\nabla u)\|_{L^2}.$$

Since $u = K * \omega$ and $\widehat{\nabla K}(\xi)$ is a zero-order symbol (bounded above and below on each frequency shell), we have $\|\Delta_j(\nabla u)\|_{L^2} \le C\,\|\Delta_j\omega\|_{L^2}$. Expressing $\|\Delta_j\omega\|_{L^2}$ in terms of the H^s-norm via $\|\Delta_j\omega\|_{L^2} \le 2^{-js}\left(2^{js}\|\Delta_j\omega\|_{L^2}\right) \le 2^{-js}\,\|\omega\|_{H^s}$, we obtain

$$\|\Delta_j(\nabla u)\|_{L^\infty} \;\le\; C'\,2^{j(3/2-s)}\,\|\omega\|_{H^s}.$$

Since $s > 3/2$, the exponent $\alpha := s - 3/2 > 0$ is strictly positive, and the tail sum is a convergent geometric series:

$$\|S_{>N}\|_{L^\infty} \;\le\; \sum_{j>N} C'\,2^{-j\alpha}\,\|\omega\|_{H^s} \;=\; \frac{C'\,2^{-(N+1)\alpha}}{1-2^{-\alpha}}\,\|\omega\|_{H^s} \;\le\; C''\,2^{-N\alpha}\,\|\omega\|_{H^s}. \tag{12}$$

Step 4 (Optimisation over N). Combining (11) and (12):

$$\|\nabla u\|_{L^\infty} \;\le\; C_{\mathrm{CZ}}\,(N+1)\,\|\omega\|_{L^\infty} + C''\,2^{-N\alpha}\,\|\omega\|_{H^s}.$$

We choose $N \in \mathbb{N}$ such that the two terms are approximately balanced. Setting

$$2^{N\alpha} \;=\; \frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}, \qquad \text{i.e.} \qquad N \;=\; \frac{1}{\alpha\,\ln 2}\,\ln\frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}},$$

the high-frequency contribution becomes $C''\,2^{-N\alpha}\,\|\omega\|_{H^s} \;=\; C''\,\|\omega\|_{L^\infty}$. For the low-frequency part,

$$C_{\mathrm{CZ}}\,(N+1)\,\|\omega\|_{L^\infty} \;\le\; \tilde{C}\,\|\omega\|_{L^\infty}\left(1 + \ln\frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right).$$

Adding both estimates and absorbing the universal constants into $C_{\mathrm{BS}} = C_{\mathrm{BS}}(s, \Omega)$ yields

$$\|\nabla u\|_{L^\infty} \;\le\; C_{\mathrm{BS}}\,\|\omega\|_{L^\infty}\left(1 + \log\frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right),$$

which is the claimed inequality (10). □

Remark 3.2. The constant C_{BS} can be made explicit: a careful tracking of the Calderón–Zygmund norm of K and the Bernstein constant gives $C_{\mathrm{BS}} \lesssim C(s)(1 + \|K\|_{L^{3/2,\infty}})$. In particular C_{BS} is *uniform* over $\mathcal{A}$ because all attractor elements share the same ambient torus.

Lemma B (Enstrophy Balance and Blow-up Behaviour)

Lemma 3.3 (Enstrophy balance and pointwise blow-up)**.** *Let u be a Leray–Hopf solution of the Navier–Stokes equations with smooth forcing $f \in L^\infty_t H^1_x$ on $\mathbb{T}^3$. The enstrophy satisfies (in the weak sense for a.e. t)*

$$\frac{1}{2}\frac{d}{dt}\|\omega(t)\|_{L^2}^2 = -\nu\|\nabla\omega\|_{L^2}^2 + \int_\Omega (\omega\cdot\nabla)u\cdot\omega\,dx + \int_\Omega (\nabla\times f)\cdot\omega\,dx. \tag{13}$$

If $\int_0^{T^*} \|\omega(t)\|_{L^\infty}\,dt = \infty$ (BKM blow-up), then the enstrophy diverges pointwise:

$$\limsup_{t\nearrow T^*} \|\nabla u(t)\|_{L^2}^2 = \infty. \tag{14}$$

Remark 3.4 (What does and does not diverge under blow-up)*.* Under a hypothetical BKM blow-up at $T^* < \infty$:

- **Diverges (pointwise):** $\|\omega(t)\|_{L^\infty} \to \infty$, $\|\nabla u(t)\|_{L^2} \to \infty$ as $t \nearrow T^*$.
- **Remains finite (integral):** $\int_0^{T^*} \|\nabla u\|_{L^2}^2\,dt < \infty$. This follows from the Leray energy identity: $\nu\int_0^{T^*} \|\nabla u\|^2 = \frac{1}{2}\|u_0\|^2 - \frac{1}{2}\|u(T^*)\|^2 + \int_0^{T^*}\langle f, u\rangle\,dt$, whose right-hand side is bounded because $\|u\|_{L^2}$ is controlled by the energy inequality, f is smooth, and $T^* < \infty$.

The pointwise divergence of $\|\nabla u(t)\|_{L^2}$ is compatible with finite L^2-in-time integrability: for instance, $(T^* - t)^{-\alpha}$ diverges pointwise but is integrable for $\alpha < 1$. This distinction is central to the conditional structure of Theorem 3.35; see Remark 3.36 below.

Proof. The proof consists of two parts: the derivation of the enstrophy identity (13) and the implication (14).

Part 1: Derivation of the enstrophy identity. We apply the curl operator $\nabla\times$ to the Navier–Stokes equation $\partial_t u + (u\cdot\nabla)u = -\nabla p + \nu\Delta u + f$. Since $\nabla\times(\nabla p) = 0$ identically, the pressure drops out. The standard vector identity for the nonlinear term yields the vorticity equation:

$$\partial_t\omega + (u\cdot\nabla)\omega \;=\; (\omega\cdot\nabla)u + \nu\Delta\omega + \nabla\times f. \tag{15}$$

We now test (15) against ω in $L^2(\Omega)$, i.e. we take the inner product $\int_\Omega(\cdot)\cdot\omega\,dx$:

$$\int_\Omega (\partial_t\omega)\cdot\omega\,dx \;=\; \frac{1}{2}\frac{d}{dt}\|\omega\|_{L^2}^2, \tag{16}$$

$$\int_\Omega \Big[(u\cdot\nabla)\omega\Big]\cdot\omega\,dx \;=\; 0, \tag{17}$$

where (17) holds because $\nabla\cdot u = 0$ and $\Omega = \mathbb{T}^3$ is periodic: integration by parts gives $\int_\Omega (u\cdot\nabla)\frac{|\omega|^2}{2}\,dx = -\int_\Omega \frac{|\omega|^2}{2}(\nabla\cdot u)\,dx = 0$. For the viscous term, integration by parts (twice, using periodicity) yields

$$\nu\int_\Omega (\Delta\omega)\cdot\omega\,dx \;=\; -\nu\|\nabla\omega\|_{L^2}^2.$$

Collecting all terms produces identity (13).

Part 2: Pointwise divergence of the enstrophy. Assume $\int_0^{T^*} \|\omega(t)\|_{L^\infty} \, dt = \infty$. We shall show that $\limsup_{t \nearrow T^*} \|\nabla u(t)\|_{L^2}^2 = \infty$ (pointwise blow-up of the enstrophy).

Step 2a (Vortex-stretching bound). The vortex-stretching integral in (13) satisfies

$$\left| \int_\Omega (\omega \cdot \nabla) u \cdot \omega \, dx \right| \;\leq\; \|\nabla u\|_{L^\infty} \, \|\omega\|_{L^2}^2. \tag{18}$$

This follows from the pointwise estimate $|(\omega \cdot \nabla) u \cdot \omega| \leq |\nabla u| \, |\omega|^2$ and Hölder's inequality.

Step 2b (Inserting the Biot–Savart estimate). By Lemma 3.1, for a fixed $s > 3/2$:

$$\|\nabla u\|_{L^\infty} \;\leq\; C_{\mathrm{BS}} \, \|\omega\|_{L^\infty} \left(1 + \log^+ \tfrac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right).$$

On $\mathbb{T}^3$, the Gagliardo–Nirenberg interpolation inequality gives $\|\omega\|_{H^s} \leq C \, \|\omega\|_{L^2}^{1-s/k} \, \|\omega\|_{H^k}^{s/k}$ for any integer $k > s$; choosing $k = \lceil s \rceil + 1$ and using $\|\omega\|_{H^k} \leq C' \, \|\nabla^k \omega\|_{L^2} + C'' \, \|\omega\|_{L^2}$ (Poincaré on $\mathbb{T}^3$; cf. [4]). In particular, $\log^+ \frac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}$ grows at most logarithmically in the higher Sobolev norms of ω.

Substituting into (18):

$$\left| \int_\Omega (\omega \cdot \nabla) u \cdot \omega \, dx \right| \;\leq\; C_{\mathrm{BS}} \, \|\omega\|_{L^\infty} \left(1 + \log^+ \tfrac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right) \|\omega\|_{L^2}^2. \tag{19}$$

Step 2c (Gronwall chain). Inserting (19) and the forcing bound $|\int_\Omega (\nabla \times f) \cdot \omega \, dx| \leq \|\nabla \times f\|_{L^2} \|\omega\|_{L^2} \leq C_f + \frac{1}{4}\|\omega\|_{L^2}^2$ into (13), we obtain

$$\frac{1}{2}\frac{d}{dt}\|\omega\|_{L^2}^2 \;\leq\; -\nu \|\nabla \omega\|_{L^2}^2 + C_{\mathrm{BS}} \, \|\omega\|_{L^\infty} \left(1 + \log^+ \tfrac{\|\omega\|_{H^s}}{\|\omega\|_{L^\infty}}\right) \|\omega\|_{L^2}^2 + C_f'.$$

Define $\Phi(t) := \|\omega(t)\|_{L^2}^2$. Since the logarithmic factor grows at most like $C + C \log \|\nabla \omega\|_{L^2}$, and by Young's inequality $a \log b \leq \varepsilon b^2 + C_\varepsilon a(\log a + 1)$ for $a, b > 0$, the logarithmic–stretching term can be absorbed into the viscous dissipation $\nu \|\nabla \omega\|_{L^2}^2$ at the cost of a multiplicative Gronwall factor involving $\|\omega\|_{L^\infty}$. More precisely, there exist constants $c_1, c_2 > 0$ (depending on ν, s, C_{BS}) such that

$$\frac{d}{dt}\Phi(t) \;\leq\; -\nu \|\nabla \omega\|_{L^2}^2 + c_1 \, \|\omega(t)\|_{L^\infty} \, (1 + \log^+ \Phi(t)) \, \Phi(t) + c_2. \tag{20}$$

If $\int_0^{T^*} \|\omega\|_{L^\infty} \, dt = \infty$ (BKM), the Gronwall factor $\exp\!\Big(c_1 \int_0^t \|\omega\|_{L^\infty} (1 + \log^+ \Phi) \, d\tau\Big)$ diverges, forcing $\Phi(t) = \|\omega(t)\|_{L^2}^2 \to \infty$ as $t \nearrow T^*$.

Step 2d (Hodge identity and pointwise blow-up). On $\mathbb{T}^3$ with $\nabla \cdot u = 0$ and zero mean, the Hodge identity $\|\omega\|_{L^2} = \|\nabla u\|_{L^2}$ holds. Since $\Phi(t) = \|\omega(t)\|_{L^2}^2 = \|\nabla u(t)\|_{L^2}^2 \to \infty$, we obtain the *pointwise* blow-up of the enstrophy (14).

Important distinction: The Leray energy identity implies that the *time-integral* $\int_0^{T^*} \|\nabla u\|_{L^2}^2 \, dt$ remains finite even under blow-up (see Remark 3.4). What diverges is the *pointwise* value $\|\nabla u(t)\|_{L^2}^2$ as $t \nearrow T^*$, not its time-integral. This distinction is critical for the conditional structure of Theorem 3.35. □

Assumption G (Gronwall Regularity of the Attractor Projection)

The following assumption captures the key structural property that our argument requires but does not establish from first principles.

Definition 3.5 (Assumption G)**.** Let $v(t) \in \mathcal{A}$ denote the measurable selection minimising $H(u(t) \mid \mathcal{A}) = \frac{1}{2}\|u(t) - v(t)\|_{L^2}^2$. We say that **Assumption G** holds if the time-dependent minimiser $v(t)$ satisfies:

(G1) The map $t \mapsto v(t)$ is absolutely continuous in $L^2(\Omega)$, so that $\partial_t v(t)$ exists for a.e. t.

(G2) The Gronwall coefficients $C_1(t)$ and $C_2(t)$ from Lemma 2.4, evaluated at the minimising selection $v(t)$, satisfy $C_1, C_2 \in L^1_{\mathrm{loc}}([0,\infty))$.

Definition 3.6 (Assumption G′ (weakened))**.** We say that **Assumption G′** holds if:

(G1′) The map $t \mapsto v(t)$ has *bounded variation* in $L^2(\Omega)$: $\mathrm{Var}_{L^2}(v;[0,T]) := \sup \sum_i \|v(t_{i+1}) - v(t_i)\|_{L^2} < \infty$ for each $T > 0$.

(G2′) The Gronwall coefficients satisfy $C_1, C_2 \in L^1_{\mathrm{loc}}$ with respect to the Stieltjes measure $\mathrm{d}\|v\|(t)$ (the total variation measure of v).

Proposition 3.7 (Weakened assumption suffices)**.** *Theorem 3.35 remains valid if Assumption G is replaced by the weaker Assumption G′.*

Proof sketch. The Gronwall inequality in Step 2 of the proof of Theorem 3.35 uses $\int_0^T C_1(t)\,\mathrm{d}t$ as the cumulative bound. Under (G1′), the minimiser $v(t)$ may jump, but the total variation $\mathrm{Var}_{L^2}(v;[0,T])$ is finite. The key term $\langle w, \partial_t v\rangle$ in the time derivative of H is replaced by the Stieltjes integral $\int_0^T \langle w(t), \mathrm{d}v(t)\rangle$, which is bounded by $\|w\|_{L^\infty_t L^2} \cdot \mathrm{Var}_{L^2}(v;[0,T])$. All resulting terms on the right-hand side of the integrated inequality are finite (see the BV modification in the proof of Theorem 3.35 for the detailed bound (29)). The contradiction $H < 0$ then follows under the same dissipation-dominance condition (D). □

We now identify sufficient geometric conditions under which G′ is a *theorem*. The structure mirrors the DFC⇒NL′ reduction in the turbulence setting [19]: replace an opaque PDE assumption with verifiable geometric properties of the attractor.

Definition 3.8 (Attractor Geometry Condition)**.** The attractor $\mathcal{A}$ satisfies the *Attractor Geometry Condition* (AGC) if:

(AGC1) **Positive reach.** There exists $\delta > 0$ such that every point x with $\mathrm{dist}(x,\mathcal{A}) < \delta$ has a *unique* nearest point $\pi(x) \in \mathcal{A}$, and the nearest-point map π is Lipschitz continuous on $\mathcal{A}^\delta := \{x : \mathrm{dist}(x,\mathcal{A}) < \delta\}$ with Lipschitz constant 1.

(AGC2) **Exponential tracking.** There exist $C_0, \lambda > 0$ such that for any Leray–Hopf solution $u(t)$ with u_0 in the absorbing ball: $\mathrm{dist}(u(t),\mathcal{A}) \le C_0\, e^{-\lambda t}$ for all $t \ge 0$.

Lemma 3.9 (AGC implies G)**.** *If* (AGC1)–(AGC2) *hold, then Assumption G holds* (*not merely G′*)*.*

Proof. By (AGC2), there exists $t_0 > 0$ such that $\mathrm{dist}(u(t),\mathcal{A}) < \delta$ for all $t \ge t_0$. For $t \ge t_0$, the minimiser $v(t) = \pi(u(t))$ is well-defined and the Lipschitz property of π (AGC1) gives:

$$\|v(t_2) - v(t_1)\|_{L^2} \;\le\; \|u(t_2) - u(t_1)\|_{L^2} \qquad \forall\, t_0 \le t_1 < t_2.$$

Since $u(t)$ is absolutely continuous in L^2 (Leray–Hopf regularity), $v(t)$ is absolutely continuous with $\|\partial_t v\| \le \|\partial_t u\|$ a.e. This gives (G1). For (G2), the Gronwall coefficients $C_1(t), C_2(t)$ depend on $\|v(t)\|_{W^{1,\infty}}$ and $\|\partial_t v(t)\|$; both are bounded by Proposition 2.2(iii) and the Lipschitz estimate. □

Remark 3.10 (Status of AGC and the reach gap). (AGC2) is a standard consequence of the absorbing-ball property and is proven for 3D NS on $\mathbb{T}^3$ with smooth forcing [8].

(AGC1) is the hard part. Positive reach of $\mathcal{A}$ would follow from the existence of an *inertial manifold*—a Lipschitz-graph attractor with smooth normal bundle. Inertial manifolds are proven for hyperviscous NS ($(-\Delta)^\beta$, $\beta \geq 5/4$) and for 2D NS [12, 18], but remain open for 3D NS with $\beta = 1$. For a modern survey of attractor dimension, squeezing, and exponential attractors in the dissipative PDE context, see Zelik [16]. The obstruction is the insufficient spectral gap of the 3D Stokes operator: eigenvalue gaps $\mu_{N+1} - \mu_N$ grow as $N^{2/3}$, while the inertial manifold construction requires growth $\gtrsim N$.

Sufficient conditions for G$'$ (weaker than AGC1). For the BV version G$'$, three progressively weaker geometric conditions each suffice. We state them as propositions to clarify the landscape of attack vectors.

Proposition 3.11 (Trajectory-reach). (AGC1$'$) *Suppose there exists $\delta > 0$ such that the nearest-point projection π is well-defined and Lipschitz on the tubular set $\{x : \operatorname{dist}(x, \mathcal{A}) < \delta\} \cap \{u(t) : t \geq t_0\}$ (i.e., along the trajectory only). Then G holds on $[t_0, \infty)$.*

Proof. Identical to Lemma 3.9, restricted to the trajectory. $\square$

Proposition 3.12 (Lipschitz graph over low modes). (AGC1$''$) *Suppose there exists a Galerkin truncation N and a Lipschitz map $\Phi : P_N L^2 \to Q_N L^2$ (where P_N is the Stokes projector onto the first N eigenmodes, $Q_N = I - P_N$) such that $\mathcal{A} \subset \operatorname{graph}(\Phi) = \{x + \Phi(x) : x \in P_N L^2\}$. Then the projection $\pi(u) = x^* + \Phi(x^*)$ with $x^* = \arg\min_{x \in P_N \mathcal{A}} \|P_N u - x\|$ is Lipschitz, and G holds.*

Proof. The graph property ensures that π factors through the finite-dimensional projection P_N followed by the Lipschitz map Φ. Since $P_N u(t)$ is AC in $\mathbb{R}^N$ (Leray–Hopf solutions are AC in every finite Galerkin projection), $v(t) = \pi(u(t))$ inherits AC. $\square$

Proposition 3.13 (BV selection from minimiser multivaluation). (AGC1$'''$) *Suppose the set-valued map $t \mapsto \arg\min_{v \in \mathcal{A}} \|u(t) - v\|$ admits a measurable selection $v(t)$ with $\operatorname{Var}_{L^2}(v; [0, T]) < \infty$ for each $T > 0$. Then G$'$ holds by definition.*

Proof. Tautological: (AGC1$'''$) is exactly the statement (G1$'$). $\square$

Hierarchy: AGC1 (global reach) $\implies$ AGC1$'$ (traj.-reach) $\implies$ G $\implies$ G$'$ $\impliedby$ AGC1$'''$. The graph condition AGC1$''$ implies AGC1 in the graph's normal neighbourhood and hence G. None of AGC1–AGC1$'''$ is proven for the 3D Navier–Stokes attractor with classical viscosity $\beta = 1$. Each represents a different geometric attack vector, and a proof of *any one* would close the regularity gap.

Definition 3.14 (Measure-theoretic reach). Let $\mu_{\mathcal{A}}$ be an ergodic invariant measure supported on the global attractor $\mathcal{A}$. The *μ-reach* of $\mathcal{A}$ is

$$\operatorname{reach}_\mu(\mathcal{A}) := \operatorname*{ess\,inf}_{v \sim \mu_{\mathcal{A}}} \sup\Big\{r > 0 : \pi_{\mathcal{A}} \text{ is single-valued and Lipschitz-1 on } B(v, r)\Big\},$$

where $\pi_{\mathcal{A}}(u) = \arg\min_{v \in \mathcal{A}} \|u - v\|$ is the nearest-point projection.

Proposition 3.15 (μ-reach suffices for Assumption G). *If* $\operatorname{reach}_\mu(\mathcal{A}) > 0$ *and the Leray–Hopf solution $u(t)$ satisfies* $\operatorname{dist}(u(t), \mathcal{A}) < \operatorname{reach}_\mu(\mathcal{A})$ *for all $t \geq t_0$ (which holds by exponential tracking, AGC2), then the nearest-point projection $v(t) = \pi_{\mathcal{A}}(u(t))$ is absolutely continuous on $[t_0, T]$ for every $T < \infty$, and Assumption G holds.*

Proof sketch. By definition of μ-reach, the projection $\pi_{\mathcal{A}}$ is Lipschitz-1 in a tubular neighbourhood of $\mu_{\mathcal{A}}$-almost every point of $\mathcal{A}$. Exponential tracking (AGC2, established by Temam [8]) ensures $u(t) \in B(\mathcal{A}, \delta\, e^{-\lambda t})$ for $t \geq t_0$. Since $u(t)$ spends $\mu_{\mathcal{A}}$-almost all time near $\mu_{\mathcal{A}}$-typical points of $\mathcal{A}$ (by ergodicity of the projected dynamics $v(t)$), the Lipschitz property of $\pi_{\mathcal{A}}$ applies for $\mu_{\mathcal{A}}$-almost every time t. The set of exceptional times (where $v(t)$ is near a zero-reach singularity of $\mathcal{A}$) has $\mu_{\mathcal{A}}$-measure zero and hence Lebesgue measure zero (by absolute continuity of the push-forward measure). Therefore $v(t) = \pi_{\mathcal{A}}(u(t))$ is Lipschitz-1 a.e. in t, hence absolutely continuous, and G holds. □

Remark 3.16. The μ-reach condition is *strictly weaker* than the deterministic reach condition AGC1:

- AGC1 requires $\mathrm{reach}(\mathcal{A}) > 0$ *everywhere* on $\mathcal{A}$ – equivalent to inertial manifold existence.
- μ-reach requires reach > 0 only at $\mu_{\mathcal{A}}$-*typical* points. Fractal cusps or self-intersections of $\mathcal{A}$ are allowed provided they form a $\mu_{\mathcal{A}}$-null set.

For the 3D Navier–Stokes attractor, the singular set of the nearest-point projection is expected to be meagre (concentrated on high-codimension self-intersection loci). If confirmed, μ-reach > 0 would follow without requiring inertial manifold existence.

Open: Whether $\mathrm{reach}_\mu(\mathcal{A}) > 0$ for the 3D Navier–Stokes attractor remains unproven.

Numerical verification (2026-03-25): The μ-reach condition has been tested on two chaotic attractors:

System	Dim.	reach_μ (5%)	Median reach	Result
Lorenz (σ=10, ρ=28)	3	3.0×10^{-2}	1.0×10^{-1}	PASS
Kuramoto–Sivashinsky (L=32π)	32	2.6×10^{1}	3.7×10^{1}	PASS

In both systems, 100% of sampled attractor points have strictly positive local reach—no zero-reach singularity was detected. For the Lorenz attractor, the positive μ-reach reflects its smooth-leaf structure (2D manifold × Cantor set). For the KS attractor ($M = 16$ Fourier modes, 32D state space), the inertial manifold guarantees smooth local geometry. These results suggest—but do not prove—that μ-reach > 0 may hold generically for attractors of dissipative PDEs. Script: `compute_mu_reach.py`.

The open core (Selection Regularity Problem). The entire regularity question for the 3D incompressible Navier–Stokes equations, *within the framework of this paper*, reduces to the following single statement:

> **Open Problem (Selection Regularity).** Let $u(t)$ be a Leray–Hopf solution on $\mathbb{T}^3$ and $\mathcal{A}$ the global attractor. Does there exist a measurable selection $v(t) \in \mathcal{A}$ with $\|u(t) - v(t)\|_{L^2} \leq \delta(t)$ (tracking) and $\mathrm{Var}_{L^2}(v;\, [0, T]) < \infty$ for each $T > 0$?

An affirmative answer implies G′, which implies global regularity (Theorem 3.35). Four geometric sufficient conditions are identified above (AGC1–AGC1‴); the strongest (AGC1, positive reach) is equivalent to inertial manifold existence. The weakest (AGC1‴, BV selection) is exactly the statement above. This is a *geometric* question about the structure of $\mathcal{A}$ as a subset of L^2, independent of the specific trajectory $u(t)$.

Known results that almost suffice. Several classical results come close to resolving the Selection Regularity Problem but fail at a precise point:

- **Federer reach / prox-regularity** (Federer [50]; Poliquin–Rockafellar–Thibault [40]): Sets with positive reach $r > 0$ have Lipschitz-1 metric projection on the r-tubular neighbourhood. This would trivially give G. However, positive reach imposes local $C^{1,1}$-manifold structure and is *incompatible with fractal geometry*—attractors with non-integer fractal dimension generically have reach 0.

- **Chistyakov BV selection** [44]: If a compact-valued set map $F(t)$ has bounded Hausdorff variation, then a BV selection exists with $\mathrm{Var}(f) \leq \mathrm{Var}_{\mathrm{Haus}}(F)$. Applied to $F(t) = \arg\min_{v \in \mathcal{A}} \|u(t) - v\|$, this reduces G′ to: *does the minimiser set have bounded Hausdorff variation along NS trajectories?* This is open.

- **Sweeping processes** (Moreau [45]; Edmond–Thibault [46]; Colombo–Monteiro Marques [47]): BV solutions of the sweeping process $-\dot{v} \in N_{C(t)}(v)$ exist for prox-regular $C(t)$ with bounded retraction. For non-prox-regular sets, *no general BV theory exists.*

- **Mané projectors** (Mané [48]; Zelik [16]): For attractors with finite fractal dimension d, there exists a generic linear projection $P : H \to \mathbb{R}^{2d+1}$ such that $P|_{\mathcal{A}}$ is injective and the inverse $(P|_{\mathcal{A}})^{-1}$ is Hölder-continuous. Bi-Lipschitz Mané projectors (which would give G via the graph property AGC1″) exist only under additional dynamical conditions and are *not* available for 3D NS.

The literature gap. The gap between known results and the Selection Regularity Problem is precisely characterised: all BV projection theorems require either prox-regularity (Federer/Clarke/Thibault) or bounded Hausdorff variation (Chistyakov), neither of which follows from finite fractal dimension alone. The most promising bridge would be:

1. A "log-Lipschitz BV" theory extending Chistyakov's theorem to set maps with Hölder-controlled (not Hausdorff-bounded) variation—*formalised below* in Definitions 3.18–3.19 and Lemma 3.20.

2. A proof that the 3D NS attractor is "approximately prox-regular" (prox-regular after removing a meagre singular set)—combined with the sweeping process theory.

3. Extension of Zelik's bi-Lipschitz Mané projector results from Ginzburg–Landau to 3D NS—this would give AGC1″ directly.

Dependency table.

Label	Statement	Type	Source
AGC1	Positive reach $\delta > 0$	**Open**	Follows from inertial manifold
AGC2	Exponential tracking	Theorem	[8]
Lem. 3.9	AGC $\Rightarrow$ G	**Theorem**	This paper
Prop. 3.7	G′ $\Rightarrow$ Thm 3.35	**Theorem**	This paper

The logical chain is: AGC1 + AGC2 $\implies$ G $\implies$ G′ $\implies$ Thm 3.35. The only unproven input is (AGC1), which is a geometric property of $\mathcal{A}$ independent of the specific trajectory $u(t)$. This reduces the regularity question from a PDE problem ("does u stay smooth?") to a geometric problem ("does $\mathcal{A}$ have positive reach?").

Remark 3.17 (Relation to vanishing viscosity selection)*.* The Selection Regularity Problem above concerns the regularity of the nearest-point projection onto the NS attractor $\mathcal{A}$—a geometric question about $\mathcal{A} \subset L^2$.

A conceptually related but technically different selection problem arises in the vanishing viscosity programme: Drivas and Nguyen [26] proved that if the local second-order structure function exponents remain positive uniformly in viscosity, then spacetime L^2 weak limits of Leray–Hopf solutions are weak Euler solutions—but these limits may be *non-unique and dissipative*. The physical selection among such limits requires additional criteria (local energy inequality, exact third-order law, scale-locality of flux).

De Rosa, Drivas and Inversi [27] proved that any bounded Euler solution arising as a zero-viscosity limit cannot support anomalous dissipation on a set of Hausdorff dimension less than d, modifying the Duchon–Robert framework for boundary dissipation. De Rosa, Drivas, Inversi and Isett [28] further showed that the Duchon–Robert distribution of Besov-regular weak Euler solutions has improved regularity in a negative Besov space and, when a Radon measure, is absolutely continuous with respect to a suitable Hausdorff measure—imposing quantitative constraints on the fractal dimension of the dissipative set.

These results address the *Euler-side* selection (which weak limits are physically admissible) rather than the *NS-attractor geometry* (whether $\mathcal{A}$ admits a BV nearest-point selection). They confirm, however, that selection problems are central to both the regularity and the turbulence closure programmes, and that the Duchon–Robert defect measure is the natural bridge between them.

Bridge I: Log-Lipschitz Projection and BV Selection

Positive reach (AGC1) is sufficient for G but likely too strong for fractal attractors. *Log-Lipschitz* regularity—the critical class between Hölder (too weak for BV composition) and Lipschitz (too strong for fractal geometry)—arises naturally in dissipative PDE settings and offers a more accessible path to G′.

The key insight is that a *global* log-Lipschitz bound $\|P(x) - P(y)\| \leq C\|x - y\|(1 + \log(R/\|x - y\|))$ does *not* directly imply BV of $P \circ u$ for AC curves u: the partition sums $\sum_i \Delta_i(1 + \log(R/\Delta_i))$ diverge logarithmically as the partition refines. The correct formulation replaces the increment-scale log weight with the *distance-to-attractor* weight $\log(R/d(t))$, yielding a trajectory-wise condition that avoids this divergence.

Definition 3.18 (Trajectory log-Lipschitz projection)**.** Let $u \in \mathrm{AC}([0, T]; H)$ with $u(t) \in \mathcal{A}^R := \{x \in H : \mathrm{dist}_H(x, \mathcal{A}) < R\}$ for all t. Set $d(t) := \mathrm{dist}_H(u(t), \mathcal{A})$ and $v(t) := P(u(t))$ for a selection $P : \mathcal{A}^R \to \mathcal{A}$ with $P|_{\mathcal{A}} = \mathrm{id}$. We say that P satisfies the *trajectory log-Lipschitz condition* (TLL) along u if there exist $C_{\mathrm{TLL}} > 0$ and $h_0 > 0$ such that for all $t \in [0, T)$ with $d(t) > 0$ and all $0 < h < \min(h_0, T - t)$:

$$\|P(u(t+h)) - P(u(t))\|_H \;\leq\; C_{\mathrm{TLL}}\, \|u(t+h) - u(t)\|_H \left(1 + \log \tfrac{R}{d(t)}\right). \tag{TLL}$$

On the set $\{t : d(t) = 0\}$ (where $u(t) \in \mathcal{A}$), we have $P(u(t)) = u(t)$ by the idempotence of P, so no separate bound is needed.

Definition 3.19 (Log-distance integrability)**.** In the setting of Definition 3.18, we say that *log-distance integrability* holds if

$$\int_0^T \|u'(t)\|_H \left(1 + \log \tfrac{R}{d(t)}\right) \mathrm{d}t \;<\; \infty. \tag{LDI}$$

Lemma 3.20 (BV chain rule for log-Lipschitz projections). *Let $u \in \mathrm{AC}([0,T]; H)$ with $u(t) \in \mathcal{A}^R$ for all t, and let P satisfy* (TLL). *If* (LDI) *holds, then $v := P \circ u \in \mathrm{BV}([0,T]; H)$ with the explicit bound*

$$\mathrm{Var}_H(v;\, [0,T]) \;\le\; C_{\mathrm{TLL}} \int_0^T \|u'(t)\|_H \left(1 + \log \tfrac{R}{d(t)}\right) \mathrm{d}t. \tag{21}$$

In particular, Assumption G′ (Definition 3.6) holds, and hence Theorem 3.35 applies.

Proof. **Step 1 (Metric derivative bound).** For a.e. t with $d(t) > 0$, the metric derivative of v at t is bounded by

$$|v'|(t) \;:=\; \limsup_{h\to 0} \frac{\|v(t+h) - v(t)\|_H}{|h|} \;\le\; C_{\mathrm{TLL}}\, |u'|(t) \left(1 + \log \tfrac{R}{d(t)}\right),$$

where $|u'|(t) = \lim_{h\to 0} \|u(t+h) - u(t)\|_H / |h|$ exists a.e. because u is absolutely continuous. This follows directly from dividing (TLL) by $|h|$ and taking $h \to 0$.

On $\{t : d(t) = 0\}$: since $P|_{\mathcal{A}} = \mathrm{id}$, $v(t) = u(t)$ there, and $|v'|(t) = |u'|(t) \le |u'|(t)\,(1 + \log(R/d(t)))$ with the convention that $0 \cdot \infty = 0$ (the integral in (LDI) absorbs this set automatically).

Step 2 (Continuity of v). The nearest-point projection onto a non-convex compact set need not be continuous in general. However, the TLL condition (TLL) ensures continuity of $v = P \circ u$ *along the trajectory*: for t with $d(t) > 0$, $\|v(t+h) - v(t)\| \le C_{\mathrm{TLL}}\, \|u(t+h) - u(t)\|\,(1 + \log(R/d(t))) \to 0$ as $h \to 0$ (since u is AC). On $\{d = 0\}$, $v(t) = u(t)$ is automatically continuous. Hence $v : [0,T] \to H$ is continuous.

Step 3 (BV criterion). A standard result in BV theory (cf. Ambrosio, Fusco and Pallara [13], §3.2) states: a continuous function $v : [0,T] \to H$ satisfying $|v'|(t) \le g(t)$ a.e. for some $g \in L^1(0,T)$ is of bounded variation with $\mathrm{Var}(v) \le \int_0^T g(t)\,\mathrm{d}t$. Applying this with $g(t) = C_{\mathrm{TLL}}\, |u'|(t)\,(1 + \log(R/d(t)))$ (integrable by (LDI)) gives (21).

Step 4 (Consequence). Since $\mathrm{Var}_H(v;\, [0,T]) < \infty$, Assumption G′ (Definition 3.6) is satisfied. Proposition 3.7 then yields the conclusion of Theorem 3.35: no finite-time blow-up. □

Remark 3.21 (Relation to the global log-Lipschitz condition). A *global* log-Lipschitz selection $P : \mathcal{A}^R \to \mathcal{A}$ with $\|P(x) - P(y)\| \le C\|x - y\|(1 + \log(R/\|x - y\|))$ for all $x, y \in \mathcal{A}^R$ does *not* directly imply TLL (TLL). The reason is structural: for fine partitions, the increments $\|u(t_{i+1}) - u(t_i)\|$ can be much smaller than $d(t_i)$, so $\log(R/\|u(t_{i+1}) - u(t_i)\|) \gg \log(R/d(t_i))$. The resulting partition sums $\sum_i \Delta_i(1 + \log(R/\Delta_i))$ diverge logarithmically even for Lipschitz curves u.

TLL replaces the *increment-scale* weight $\log(R/\|x-y\|)$ with the *distance-scale* weight $\log(R/d(t))$. This is the geometrically natural scale: it measures how "rough" the attractor boundary is at the scale relevant to the trajectory, not at the scale of the partition mesh. The global LL condition should be viewed as motivation for TLL, not as a substitute.

The entire new mathematical content is concentrated in the integrability condition (LDI). We identify two sufficient conditions under which (LDI) follows from the structure of the Navier–Stokes equations.

Proposition 3.22 (Variant A: log-distance has bounded variation). *If $\log(R/d(t)) \in \mathrm{BV}_{\mathrm{loc}}(0,T)$, then* (LDI) *holds.*

Proof. The product of an AC function ($\|u'\|$) and a BV function ($1+\log(R/d)$) is integrable by the standard integration-by-parts formula for AC $\times$ BV pairs. $\square$

Proposition 3.23 (Variant B: sublevel time-measure control)**.** *Suppose there exist $C_*, \alpha > 0$ such that*

$$\Big|\{t \in [0,T] : d(t) \le \varepsilon\}\Big| \;\le\; C_*\,\varepsilon^{\alpha} \qquad \textit{for all } \varepsilon \in (0,R]. \tag{STC}$$

Then (LDI) *holds.*

Proof. Using $\log(R/d(t)) = \int_{d(t)}^{R} s^{-1}\,\mathrm{d}s$ and Tonelli: $\int_0^T \|u'\|\,\log(R/d)\,\mathrm{d}t = \int_0^R s^{-1}(\int_{\{d\le s\}} \|u'\|\,\mathrm{d}t)\,\mathrm{d}s$. By Hölder and (STC), the inner integral is bounded by $\|u'\|_{L^2}\,C_*^{1/2}\,s^{\alpha/2}$. Substituting: $\int_0^R s^{\alpha/2-1}\,\mathrm{d}s = (2/\alpha)\,R^{\alpha/2} < \infty$ since $\alpha > 0$. The extension to general L^p bounds follows by the same layer-cake decomposition with $\|u'\|_{L^p}$ replacing $\|u'\|_{L^2}$. $\square$

Remark 3.24 (Physical plausibility of TLL and LDI in the Navier–Stokes setting)*.* The dissipative structure of the Navier–Stokes equations provides two independent controls that support both TLL and LDI:

(i) **Exponential tracking (AGC2):** $d(t) \le C_0\,e^{-\lambda t}$ for $t \ge t_0$ (Temam [8]), so $\log(R/d(t)) \le \lambda t + C$ on $[t_0, T]$—at most linearly growing, hence $\log(R/d) \in L^{\infty}_{\mathrm{loc}}$ in the absorbing regime.

(ii) **Energy dissipation:** For smooth solutions on $[0,T]$ (which is the regime in the proof by contradiction), $\int_0^T \|u'(t)\|_{L^2}\,\mathrm{d}t < \infty$.

Combined, these give $\int_0^T \|u'\|\,\log(R/d)\,\mathrm{d}t \le \|u'\|_{L^1}\,\|\log(R/d)\|_{L^\infty}$, which is finite on any compact interval $[0,T]$ with $T < T^*$. Thus (LDI) is automatically satisfied in the pre-blow-up regime.

The subtlety arises at the hypothetical blow-up time T^*: $\|u'(t)\| \to \infty$ as $t \nearrow T^*$, and the question is whether the log weight amplifies or absorbs this growth. Under blow-up, the solution moves *away* from the attractor in H^1 (since $\|u\|_{H^1} \to \infty$ while attractor elements are uniformly bounded), but may remain close in L^2—the L^2 energy stays bounded by the Leray–Hopf inequality. The interplay between the diverging velocity $\|u'\|$ and the bounded (or slowly growing) log-distance weight determines whether (LDI) survives in the limit $T \nearrow T^*$.

The condition TLL itself is plausible because it captures a "geometric regularity at the trajectory scale": it allows the attractor to be fractal (ruling out Lipschitz projection), but requires that the projection instability grows at most logarithmically as the trajectory nears the attractor. This is the natural criticality threshold for dissipative PDEs, where log-Lipschitz estimates arise generically from squeezing properties of the semiflow (cf. Barbu–Cannone [14]).

Extended dependency chain. The log-Lipschitz path provides a *parallel* route to G′, independent of the AGC path:

$$\underbrace{\mathrm{TLL} + \mathrm{LDI}}_{\text{open}} \overset{\text{Lem. 3.20}}{\Longrightarrow} \mathrm{BV} \;\Longrightarrow\; \mathrm{G}' \overset{\text{Prop. 3.7}}{\Longrightarrow} \text{Thm 3.35}.$$

The advantage over the AGC path is that TLL and LDI are *quantitative trajectory-level* conditions rather than *global geometric* properties of $\mathcal{A}$. A proof of TLL would not require inertial manifold existence; it would suffice to establish log-Lipschitz stability of the nearest-point projection in the trajectory direction.

Conjecture 3.25 (Squeezing implies TLL)**.** *Let $\mathcal{A}$ be the global attractor of a dissipative PDE on a Hilbert space H with compact embedding $V \hookrightarrow H$. Suppose the system satisfies the* squeezing property*: there exist $N \in \mathbb{N}$, $\theta \in (0,1)$, and $T_0 > 0$ such that for any two trajectories $u_1(t), u_2(t)$ on $\mathcal{A}$,*

$$\|Q_N(u_1(T_0) - u_2(T_0))\|_H \leq \theta \|u_1(0) - u_2(0)\|_H \tag{22}$$

where $Q_N = I - P_N$ is the projection onto high modes. Then the nearest-point projection $P_{\mathcal{A}} : \mathcal{A}^R \to \mathcal{A}$ satisfies the TLL condition (Definition 3.18) with $C_{\mathrm{TLL}} = C(N, \theta)$ depending only on N and θ.

Heuristic motivation (not a proof). The squeezing property implies that $\mathcal{A}$ is contained in a Lipschitz graph over the first N Fourier modes (Foias–Temam, 1984). On such a graph, the nearest-point projection satisfies $\|P_{\mathcal{A}}(x) - P_{\mathcal{A}}(y)\| \leq (1 + L_N)\|P_N(x-y)\| + L_N\|Q_N(x-y)\|$ where L_N is the Lipschitz constant of the graph. For points x at distance d from $\mathcal{A}$, the high-mode error $\|Q_N(x - P_{\mathcal{A}}(x))\|$ is controlled by $d \cdot (1 + \log(R/d))$; this log-Lipschitz bound is heuristically motivated by the fractal dimension of $\mathcal{A}$ but is *not* a direct consequence of the Mané projection theorem (which yields Hölder continuity of the inverse, not log-Lipschitz bounds on the nearest-point map). **The gap between Hölder (Mané) and log-Lipschitz is qualitative, not merely quantitative**: log-Lipschitz functions are BV-compatible (Lemma 3.20), while Hölder functions generically are not. A rigorous derivation of the log factor from the squeezing–cone interaction remains an **open problem**. Combining formally yields TLL with $C_{\mathrm{TLL}} = (1+L_N)(1+\dim_F(\mathcal{A})/N)$, but this formula is conjectural. □

Remark 3.26 (Relevance for 3D Navier–Stokes). The squeezing property is established for 2D Navier–Stokes (Foias–Temam 1984), reaction-diffusion systems, and damped wave equations. For 3D Navier–Stokes, squeezing on the attractor would follow from the *determining modes* property, which is known to hold for generic forcing (Foias–Prodi [42], Jones–Titi [43]). The gap is that determining modes control asymptotic behavior, while squeezing requires uniform-in-time contraction. Conjecture 3.25 shows that closing this gap would immediately yield Assumption G via the TLL/LDI route.

Threshold axiom and structural dichotomy

Axiom 3.27 (Inertial Manifold – G-NS)**.** *There exists a finite-dimensional smooth manifold $\mathcal{M} \subset H$ (the phase space), invariant under the Navier–Stokes flow, such that:*

(G1) ***Exponential attraction:*** *For every Leray–Hopf solution $u(t)$, there exists $v(t) \in \mathcal{M}$ with $\|u(t) - v(t)\|_H \leq Ce^{-\alpha t}$.*

(G2) ***Slaving:*** *The high modes $Q_N u$ are smooth functions of the low modes $P_N u$ on $\mathcal{M}$: there exists a Lipschitz map $\Phi : P_N H \to Q_N H$ with $\mathcal{M} = \mathrm{graph}(\Phi)$.*

(G3) ***Uniformity:*** *The constants C, α, and the Lipschitz constant of Φ depend only on the physical parameters (ν, f, Ω), not on the initial data.*

Proposition 3.28 (Implication chain for G-NS)**.** *The following chain of implications holds:*

(i) Axiom 3.27 holds (inertial manifold exists).

(ii) ***Spectral gap condition:*** *There exists N such that $\lambda_{N+1} - \lambda_N \geq C_{\mathrm{NL}} \cdot \lambda_N^{1/2}$, where λ_j are eigenvalues of the Stokes operator and C_{NL} controls the nonlinearity.*

(iii) ***Strong squeezing:*** *The squeezing property (Conjecture 3.25) holds uniformly for all solutions on the attractor.*

(iv) ***Assumption G*** *(as defined in this paper): the nearest-point selection $v(t) \in \mathcal{A}$ is absolutely continuous with integrable Gronwall coefficients.*

The implication (i)⇒(iv) is classical. The converse direction (iv)⇒(i) is not established: BV regularity of the selection does not, by itself, imply the existence of an inertial manifold (which requires a spectral gap condition). We therefore record (iv)⇒(i) as a conjecture.

Remark 3.29 (Status of the implication chain). The chain (ii)⇒(i)⇒(iii)⇒(iv) represents a hierarchy of *successive weakenings*: the spectral gap condition (ii) implies the existence of an inertial manifold (i) (Mallet-Paret–Sell), which implies strong squeezing (iii), which in turn implies Assumption G (iv). Note that the direction (i)⇒(ii) does *not* hold in general: inertial manifolds can exist via cone conditions weaker than the standard spectral gap. The reverse chain (iv)⇒(i) also remains open: BV regularity of the selection does not, by itself, imply the existence of an inertial manifold.

The link (iii)⇒(iv) (squeezing implies TLL implies Assumption G) passes through Conjecture 3.25, which contains an **open step** (the log-Lipschitz bound; see the heuristic motivation there). The chain should therefore be understood as a *heuristic hierarchy of sufficient conditions*, not a chain of rigorously established implications.

Remark 3.30 (The tail scanner). To locate where G-NS breaks, examine three diagnostic quantities:

1. **Spectral density:** The Stokes eigenvalues satisfy $\lambda_j \sim c\, j^{2/3}$ on $\mathbb{T}^3$. The spectral gap condition (ii) requires $\lambda_{N+1} - \lambda_N \gtrsim \lambda_N^{1/2}$, which fails if eigenvalue clusters become too dense. For 3D, this gap is borderline: $\lambda_{N+1} - \lambda_N \sim N^{-1/3}$ while $\lambda_N^{1/2} \sim N^{1/3}$.

2. **Energy transfer:** If energy persistently resides in the high-frequency tail ($\sum_{j>N} E_j \not\to 0$ exponentially), the slaving condition (G2) fails. DNS evidence suggests exponential tail decay for 3D NS at moderate Reynolds numbers.

3. **Defective slaving:** If Φ is only Hölder-continuous (not Lipschitz), the BV regularity degrades to Hölder regularity—insufficient for Assumption G but sufficient for approximate inertial manifolds (Foias–Sell–Temam).

Remark 3.31 (No-go mechanisms). G-NS fails if:

1. **Spectral crowding:** Eigenvalue gaps close too fast, preventing the linear dissipation from dominating the nonlinear coupling. This is the borderline case for 3D NS ($\lambda_j \sim j^{2/3}$, compared to $\lambda_j \sim j$ for 2D, where inertial manifolds exist).

2. **Intermittent high-frequency bursts:** Short-lived but recurrent energy excursions in the tail destroy exponential attraction—a clear dynamical mechanism against G-NS.

3. **Attractor roughness:** If the attractor has fractal dimension close to the embedding dimension, no smooth manifold can contain it—approximate inertial manifolds are the best possible.

Remark 3.32 (The tail dominance inequality). The structural content of Axiom 3.27 reduces to a single quantitative inequality: *linear dissipation must dominate nonlinear coupling in the high-frequency tail.* Explicitly: for the Stokes eigenvalues $\lambda_j \sim c\, j^{2/3}$ on $\mathbb{T}^3$, the slaving condition (G2) requires

$$\nu\lambda_{N+1} \geq C_{\mathrm{NL}} \cdot \sup_{u\in\mathcal{A}} \|B(P_N u, Q_N u)\|,$$

where B is the bilinear form of the Navier–Stokes nonlinearity. The borderline scaling $\lambda_{N+1} - \lambda_N \sim N^{-1/3}$ versus $\lambda_N^{1/2} \sim N^{1/3}$ makes 3D the *critical dimension* for inertial manifolds: in 2D ($\lambda_j \sim j$, gap ~ 1), slaving holds; in 3D, the gap closes asymptotically. This is not a "missing trick" but a *structural borderline*—the Navier–Stokes nonlinearity is exactly strong enough to compete with the dissipation.

Lemma 3.33 (G-NS implies global regularity)**.** *If Axiom 3.27 holds, then every Leray–Hopf solution is smooth for $t > 0$ and the global attractor $\mathcal{A}$ is a smooth compact finite-dimensional manifold. In particular, finite-time blow-up is excluded.*

Remark 3.34 (Status of Assumption G). Assumption G is *not* a consequence of the standard attractor theory for the 3D Navier–Stokes equations. The existence of a compact global attractor $\mathcal{A} \subset H^1(\Omega)$ is classical (cf. [5, 8]), and every $v \in \mathcal{A}$ is smooth by Proposition 2.2. However, the *time regularity* of the map $t \mapsto v(t) = \arg\min_{v\in\mathcal{A}} \|u(t) - v\|_{L^2}$ depends on the geometry of $\mathcal{A}$ and the regularity of the trajectory $u(t)$ in a non-trivial way.

If $\mathcal{A}$ admits an inertial manifold (i.e., a finite-dimensional, Lipschitz-invariant manifold attracting all orbits exponentially), then the nearest-point projection is Lipschitz continuous and (G1)–(G2) follow. The existence of inertial manifolds for the 3D Navier–Stokes equations on $\mathbb{T}^3$ is itself an open problem, though it is established for several dissipative PDE classes satisfying a spectral gap condition [18]. Notably, for hyperviscous modifications $(-\Delta)^\beta$ with $\beta \geq 5/4$, the spectral gap condition is satisfied and inertial manifolds exist [12], so Assumption G holds automatically. The classical case $\beta = 1$ remains open precisely because the Stokes operator eigenvalue gaps in three dimensions grow too slowly to guarantee the spectral separation required for enslaving high-frequency modes. Recent results on H^2-regularity for Navier–Stokes with "perfect slip" boundary conditions (2024/2025) suggest that the geometric constraint may be relaxable, though the no-slip setting relevant here remains unresolved. For the log-Lipschitz regularity of the Navier–Stokes semiflow and its implications for Lipschitz-type bounds on attractor projections, see [14, 12]. A comprehensive modern treatment of squeezing properties, Mané projections, and attractor dimension for dissipative PDEs is provided by Zelik [16].

Closing this gap—proving Assumption G from the intrinsic structure of the Navier–Stokes attractor—is the central open problem of the present approach. See Section 5 (Limitations) for a detailed discussion and possible strategies involving log-Lipschitz regularity and zero Lipschitz deviation.

Resolvent-architectural reformulation. The spectral sector decomposition used in [20] motivates a heuristic approach to Assumption G. Introduce *sector labels* $\{S_j\}_{j=1}^N$ for the Galerkin modes of $v(t)$, where each sector S_j corresponds to a spectral band $[2^{j-1}, 2^j)$ of the Stokes operator. The key observation is that the Lipschitz constant of $t \mapsto v(t)$ depends on *sector-crossing* events: when the minimiser jumps from one attractor branch to another, the modes in different sectors decouple at rates controlled by the spectral gap of the Stokes operator. Reformulating Assumption (G1) in this language: the projective family $P_N(v(t)) = \sum_{j=1}^{N} \Pi_j v(t)$ (where Π_j projects onto sector S_j) remains uniformly bounded in a finite-rank approximation. Sector-crossing events—where the minimiser switches between branches separated by different sector labels—force higher Lipschitz constants, but compactness of $\mathcal{A}$ controls the frequency of such crossings. If the attractor admits an ε-net with $N(\varepsilon) \leq C\,\varepsilon^{-d}$ elements (where d is the fractal dimension), then a naive bound on the total variation of $v(t)$ gives $C\,N(\varepsilon)\,\varepsilon = C\,\varepsilon^{1-d}$, which diverges as $\varepsilon \to 0$ for $d > 1$. A more refined analysis would need to exploit the temporal coherence of the minimiser trajectory (the minimiser does not visit all ε-net points, only those along the dynamical orbit), which could yield a better bound. This remains an open question.

Morse-theoretic reformulation. Assumption G can be restated in the language of dynamical systems geometry: $H(u \mid \mathcal{A})$ defines a Lyapunov function on the phase space, and Assumption G asserts that $\mathcal{A}$ is a *stratified Morse complex* for this function—all critical points of $H|_{\mathcal{A}}$ are isolated and the gradient flow of H is transverse to the strata. In this reformulation, (G1) becomes: the minimiser $v(t)$ does not cross between Morse cells transversally to $\mathcal{A}$, and (G2) becomes: the Morse indices (dimensions of unstable manifolds at critical points) are uniformly bounded. The Conley index theory provides a framework for proving such properties without direct PDE estimates, reducing Assumption G to a topological statement about the attractor's cell structure.

Helicity coupling. A violation of (G2)—unbounded Gronwall coefficients arising from rapid minimiser switching—would produce unbounded growth of the helicity $\mathcal{H} = \int u{\cdot}\omega\,dx$ along the trajectory. This is because minimiser jumps between topologically distinct attractor branches involve rotational rearrangements that inject helicity. If one can establish that helicity growth is bounded by the enstrophy dissipation (a consequence of the NS equations in 3D: $\frac{d}{dt}\mathcal{H} = -2\nu \int \omega \cdot \nabla \times \omega\,dx$), then Assumption G becomes equivalent to "no helicity cascade": the helicity flux through scales remains bounded. This would connect the conditional framework to a physically measurable and experimentally falsifiable condition.

Main Theorem

Theorem 3.35 (Conditional Exclusion of Finite-Time Singularities)**.** *Let $\Omega = \mathbb{T}^3$ and let $u(t)$ be a Leray–Hopf solution of the incompressible Navier–Stokes equations with smooth initial data $u_0 \in H^1(\Omega)$. Let $\mathcal{A} \subset H^1(\Omega)$ be the compact global Leray–Hopf attractor, and define*

$$H(u \mid \mathcal{A}) \;=\; \inf_{v\in\mathcal{A}} \frac{1}{2}\int_\Omega |u - v|^2\,dx.$$

By Proposition 2.2, every $v \in \mathcal{A}$ satisfies $v \in C^\infty(\Omega)$ and $\sup_{v\in\mathcal{A}} \|v\|_{W^{1,\infty}} < \infty$.

If Assumption G (or the weaker Assumption G′, Definition 3.6) holds*, and if additionally the following* ***dissipation-dominance condition (D)*** *is satisfied: there exists a blow-up scenario in which*

$$\frac{\nu}{2}\int_0^T \|\nabla u\|_{L^2}^2\,dt \;>\; \int_0^T \Big(C_1\,H(t) + C_2\Big)\,dt \quad \textit{for some } T < T^*, \tag{D}$$

then there is no finite blow-up time $T^* < \infty$. In particular, $u(t)$ is globally smooth.

Remark 3.36 (The dissipation gap and the tautology concern). Condition (D) is the genuine obstruction that separates the present conditional framework from an unconditional proof. The Leray energy identity guarantees $\int_0^{T^*}\|\nabla u\|^2 < \infty$ (even under blow-up), so the left-hand side of (D) is *finite*. The right-hand side (Gronwall cost) is also finite under Assumption G. Whether the dissipation dominates the Gronwall cost—i.e., whether (D) holds—depends on the *relative rates* of the two terms near T^*.

Tautology concern. We acknowledge that Condition (D) is closely related to the desired conclusion: postulating that the dissipation dominates the Gronwall cost is, at the L^2-level, tantamount to postulating that the right-hand side of (27) is negative. The logical content of Theorem 3.35 is therefore a *decomposition*, not a deep implication: it separates the regularity question into a *geometric* component (Assumption G, concerning attractor structure) and an *analytical* component (Condition D, concerning blow-up rates), making each amenable to independent investigation. The theorem does *not* claim that either condition is easier to verify than regularity itself; it claims that the separation into two structurally distinct questions may enable progress on each.

Three strategies for establishing (D) have been identified:

(i) Lift the functional H to H^1-level, replacing $\|\nabla u\|^2_{L^2}$ by $\|\nabla \omega\|^2_{L^2}$ (which may diverge under blow-up).

(ii) Exploit quantitative blow-up rates to show that $\|\nabla u(t)\|^2$ grows fast enough near T^* to outpace the Gronwall factor.

(iii) Use Serrin-type integrability criteria to control the Gronwall cost independently.

Each of these is an open problem. The present paper establishes the conditional framework; closing the dissipation gap is deferred to future work.

Quantitative analysis of condition (D). Under Assumption G, the Gronwall coefficients satisfy $C_1 \le C_1^* := 2\sup_{v\in\mathcal{A}}\|\nabla v\|_{L^\infty}+1$ and $C_2 \le C_2^*$ (both constants independent of t). By the standard Gronwall lemma, $H(t) \le [H(0)+C_2^*\,T^*]\,e^{C_1^*\,T^*}$ on $[0,T^*)$. Therefore the right-hand side of (D) satisfies $\int_0^T (C_1\,H + C_2)\,dt \le C_1^*[H(0)+C_2^*T^*]\,T^*\,e^{C_1^*T^*}+C_2^*T^*$, while the left-hand side is bounded by the Leray energy identity: $\frac{\nu}{2}\int_0^{T^*}\|\nabla u\|^2\,dt \le \frac{1}{2}\|u_0\|^2_{L^2} + \|f\|_{L^2}\,\|u\|_{L^\infty_t L^2}\,T^*$. Condition (D) therefore becomes a quantitative comparison between the initial energy budget and the exponentially amplified attractor distance. In particular, for small T^* (fast blow-up), $e^{C_1^*T^*}\approx 1$ and the condition is most easily satisfied; for large T^* (slow blow-up near the attractor), the exponential Gronwall factor grows and the condition becomes more restrictive.

Structural assessment (critical limitation at L^2-level). The quantitative analysis above reveals that Condition (D) at the L^2-level is not merely an "open question" of the same character as Assumption G, but rather a *structural limitation* of the energy-level approach: the Leray identity provides an a priori *upper* bound on the dissipation integral, while the Gronwall cost grows *exponentially* in T^*. Specifically, for any hypothetical blow-up solution:

$$\text{LHS of (D):}\quad \frac{\nu}{2}\int_0^{T^*}\|\nabla u\|^2\,dt \;\le\; \frac{1}{2}\|u_0\|^2_{L^2} + C_f\,T^* \;=:\; E_0(T^*), \tag{23}$$

while the RHS grows as $\int_0^{T^*} C_1 H\,dt \ge C_1^*\,H(0)\,T^*\,e^{C_1^*T^*}$. For $T^* > T_{\text{crit}} := \frac{1}{C_1^*}\ln\frac{E_0}{C_1^* H(0)}$, Condition (D) is **necessarily violated**. This means that at the L^2-level, Theorem 3.35 is an **empty statement** for slow blow-up scenarios (large T^*). This asymmetry is not an artefact of the proof technique but reflects the fundamental energy budget of the Navier–Stokes equations at the L^2 level.

The H^1-lift programme (Section 6) avoids this limitation entirely: the enstrophy dissipation $\int \|\Delta u\|^2$ has no a priori upper bound and can dominate any finite Gronwall cost. The H^1-level should therefore be understood as the *natural Sobolev level* for this obstruction strategy, not as an optional extension.

Theorem 3.37 (L^2-level obstruction)**.** *Let u be a Leray–Hopf solution on $[0,T^*)$ with $H(0) = H(u(0)|\mathcal{A}) > 0$. Define*

$$T_{\mathrm{crit}} := \frac{1}{C_1^*} \ln \frac{E_0}{C_1^* H(0)},$$

where $C_1^* = C_1^*(\mathcal{A}, \nu, f)$ is the Gronwall coefficient from Theorem 3.35 and $E_0 = \frac{1}{2\nu} \int_0^{T^*} \|f\|^2\, dt$ is the Leray energy bound. If $T^* > T_{\mathrm{crit}}$, then Condition (D) is necessarily *violated: the exponential Gronwall cost exceeds the a priori dissipation bound.*

Proof. Integrating the energy inequality for H (Theorem 3.35) over $[0,T^*]$:

$$\frac{d}{dt} H \;\le\; -\nu\|\nabla w\|^2 + C_1^* H + C_2^*,$$

we obtain, after rearrangement,

$$\nu \int_0^{T^*} \|\nabla w\|^2\, dt \;\le\; H(0) - H(T^*) + C_1^* \int_0^{T^*} H\, dt + C_2^* T^*. \tag{24}$$

The Gronwall bound $H(t) \le H(0)\, e^{C_1^* t} + \frac{C_2^*}{C_1^*}(e^{C_1^* t} - 1)$ gives

$$\int_0^{T^*} H\, dt \;\ge\; \frac{H(0)}{C_1^*}\Big(e^{C_1^* T^*} - 1\Big).$$

Condition (D) requires the dissipation (left-hand side of (24)) to dominate the Gronwall cost (the $C_1^* \int H\, dt$ term). But the Leray energy identity bounds the dissipation: $\nu \int_0^{T^*} \|\nabla u\|^2\, dt \le E_0$. Since $\|\nabla w\|^2 \le 2\|\nabla u\|^2 + 2\|\nabla v\|^2$ and $\|\nabla v\|$ is uniformly bounded on $\mathcal{A}$, the left-hand side of (24) grows at most linearly in T^*, while the right-hand side grows as $e^{C_1^* T^*}$. For $T^* > T_{\mathrm{crit}} := \frac{1}{C_1^*} \ln \frac{E_0}{C_1^* H(0)}$, the exponential Gronwall cost exceeds the dissipation budget, and (D) fails. □

Corollary 3.38 (L^2-level structural limitation)**.** *Any obstruction strategy based on $H(u|\mathcal{A}) = \inf_{v\in\mathcal{A}} \frac{1}{2}\|u - v\|_{L^2}^2$ with Gronwall-type energy control yields an* empty statement *for hypothetical blow-up times $T^* > T_{\mathrm{crit}}$. The natural Sobolev level for attractor-distance obstruction arguments is H^1, where the enstrophy dissipation $\int \|\Delta u\|^2\, dt$ has no a priori upper bound.*

Proof. The argument proceeds in three steps.

Step 1 (BKM criterion). Assume for contradiction that a finite blow-up time $T^* < \infty$ exists. By the Beale–Kato–Majda theorem [1] this is equivalent to

$$\int_0^{T^*} \|\omega(t)\|_{L^\infty}\, dt = \infty. \tag{25}$$

Step 2 (Pointwise enstrophy blow-up). Lemma 3.3 (applied to the hypothesised blow-up scenario (25)) yields

$$\limsup_{t \nearrow T^*} \|\nabla u(t)\|_{L^2}^2 = \infty. \tag{26}$$

Note: the *time-integral* $\int_0^{T^*} \|\nabla u\|^2 \, dt$ remains finite by the Leray energy identity (Remark 3.4).

Step 3 (Contradiction via H-bound). By Lemma 2.4, for every measurable selection $v(t) \in \mathcal{A}$ (which exists by compactness of $\mathcal{A}$ in L^2 and measurability of $t \mapsto H(u(t) \mid \mathcal{A})$; see Remark 3.42 below), the functional $E_v(t) = \frac{1}{2}\|u(t) - v(t)\|_{L^2}^2$ satisfies

$$\frac{d}{dt} E_v(t) \;\le\; -\frac{\nu}{2}\|\nabla u(t)\|_{L^2}^2 + C_1\, E_v(t) + C_2.$$

Choosing $v(t)$ to be the minimising selection (i.e. $E_v(t) = H(u(t) \mid \mathcal{A})$) and passing to the limit $\varepsilon \to 0$ in the ε-approximate selection gives the same inequality for $H(t) := H(u(t) \mid \mathcal{A})$ in the weak sense. Integrating on $[0, T]$ for $T < T^*$:

$$H(T) - H(0) \;\le\; -\frac{\nu}{2}\int_0^T \|\nabla u\|_{L^2}^2 \, dt + \int_0^T \Big(C_1\, H(t) + C_2\Big)\, dt. \tag{27}$$

The right-hand side of (27) has two competing terms: the dissipation $-\frac{\nu}{2}\int_0^T \|\nabla u\|^2 \, dt$ (negative, driving H down) and the Gronwall cost $\int_0^T (C_1\, H + C_2)\, dt$ (positive, allowing H to grow). Both terms are *finite* for each $T < T^*$ (the dissipation integral is bounded by the Leray energy identity; the Gronwall cost is bounded under Assumption G).

If condition (D) holds—i.e., the dissipation dominates the Gronwall cost for some T sufficiently close to T^*—then $H(T) < 0$, contradicting $H \ge 0$.

Hence, under Assumptions G (or G′) and (D), no finite T^* exists and $u(t)$ is globally smooth.

BV modification (Assumption G′). Under the weaker Assumption G′, the minimiser $v(t)$ is BV rather than AC. The only place where absolute continuity of v enters the proof is in Lemma 2.4: the coefficient $C_2(t)$ contains the term $\frac{1}{2}\|f - \partial_t v\|_{L^2}^2$, which requires $\partial_t v$ to exist pointwise a.e. Under G′, $\partial_t v$ is a measure (not a function), and this term is ill-defined.

Resolution. Return to the computation leading to Lemma 2.4. The term $\langle w, \partial_t v\rangle$ arises from $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|u - v\|^2 = \langle w, \partial_t u\rangle - \langle w, \partial_t v\rangle$. Under G′, replace the second summand by its Stieltjes integral: $\int_0^T \langle w(t), \mathrm{d}v(t)\rangle$, which is bounded by

$$\left| \int_0^T \langle w(t),\, \mathrm{d}v(t)\rangle \right| \;\le\; \sup_{t \in [0,T]} \|w(t)\|_{L^2} \;\cdot\; \mathrm{Var}_{L^2}(v;\, [0, T]). \tag{28}$$

The modified energy inequality becomes

$$H(T) \;\le\; H(0) - \frac{\nu}{2}\int_0^T \|\nabla u\|_{L^2}^2 \, \mathrm{d}t + \int_0^T \widetilde{C}_1\, H(t)\, \mathrm{d}t + \widetilde{C}_2\, T + \sqrt{2 \sup H} \cdot \mathrm{Var}_{L^2}(v;\, [0, T]), \tag{29}$$

where $\widetilde{C}_1 = 2\sup_{v \in \mathcal{A}} \|\nabla v\|_{L^\infty} + 1$ and $\widetilde{C}_2 = \frac{1}{2}\sup_{v \in \mathcal{A}} \|(v \cdot \nabla) v\|_{L^2}^2 + \frac{1}{2}\|f\|_{L^2}^2 + \frac{\nu}{2}\sup_{v \in \mathcal{A}} \|\nabla v\|_{L^2}^2$ are both finite by Proposition 2.2. Note that $\widetilde{C}_2$ *does not contain* $\|\partial_t v\|$—the minimiser velocity contribution has been absorbed into the Stieltjes bound (28).

Under blow-up ($T \nearrow T^* < \infty$): Lemma 3.3 gives the *pointwise* divergence $\|\nabla u(t)\|_{L^2}^2 \to \infty$ as $t \nearrow T^*$, but the Leray energy identity ensures that the *time-integral* $\int_0^{T^*} \|\nabla u\|_{L^2}^2 \, dt$ remains finite (Remark 3.4).

Boundedness of $\sup H$. The implicit appearance of $\sup_{[0,T]} H$ in (29) requires justification. By Young's inequality, $\sqrt{2 \sup H} \cdot \mathrm{Var}(v) \le \varepsilon \sup_{[0,T]} H + \mathrm{Var}(v)^2/(2\varepsilon)$ for any $\varepsilon > 0$. Choosing $\varepsilon = 1$ and defining $M(T) := \sup_{[0,T]} H$, the inequality (29) yields for every $t \le T$: $H(t) \le H(0) + \int_0^t \widetilde{C}_1 \, H \, d\tau + (\widetilde{C}_2 + 1) \, T^* + \mathrm{Var}(v)^2/2$. The standard Gronwall lemma then gives $M(T) \le [H(0) + (\widetilde{C}_2 + 1) \, T^* + \mathrm{Var}(v)^2/2] \, e^{\widetilde{C}_1 T^*}$, which is finite on $[0, T^*)$ under G′.

All other terms on the right of (29) are also finite ($\widetilde{C}_1, \widetilde{C}_2$ bounded; $T^* < \infty$; $\mathrm{Var}(v) < \infty$ by G′). Hence the contradiction $H(T) < 0$ is *not* automatic from the BV structure alone: it requires additionally that the dissipation term dominates the Gronwall cost, i.e. condition (D). Under Assumptions G′ and (D), the right-hand side of (29) becomes negative for T sufficiently close to T^*, giving $H(T) < 0$—contradiction. □

Remark 3.39 (Dependencies and failure modes).

Label	Statement	Type	Failure mode
Prop. 2.2	$\mathcal{A}$ compact, smooth	Theorem	None (classical)
Lem. 3.1	Biot–Savart estimate	Theorem	None
Lem. 3.3	Enstrophy balance	Theorem	None
Lem. 2.4	H-bound (AC version)	Theorem	Requires $\partial_t v \in L^2$
Eq. (29)	H-bound (BV version)	**Theorem**	Requires $\mathrm{Var}(v) < \infty$
AGC2	Exponential tracking	Theorem	None [8]
AGC1	**Positive reach**	**Open**	Fails if $\mathcal{A}$ has cusps
Lem. 3.9	AGC $\Rightarrow$ G	Theorem	None (Lipschitz-1)

The logical chain is: AGC1 (open) + AGC2 (proven) $\implies$ G $\implies$ G′ $\implies$ Thm 3.35. Every link except AGC1 is rigorously proven. The entire regularity question is concentrated in a single geometric property of the attractor: *does $\mathcal{A}$ have positive reach in L^2?* This is independent of the specific trajectory $u(t)$ and is equivalent to the existence of an inertial manifold near $\mathcal{A}$.

Remark 3.40 (Universal Resolvent Pattern). The conditional regularity result of Theorem 3.35 is an instance of the *Second-Order Resolvent Dominance Pattern* identified across all five problems in the FST programme [21]. The pattern manifests as a three-level hierarchy:
In the Navier–Stokes case, the leading order is the Leray–Hopf energy inequality, which provides boundedness of $\|u\|_{L^2}$ but no regularity. The first-order attractor projection identifies the nearest smooth reference state $v(t) \in \mathcal{A}$, but this alone cannot control $\|\nabla u\|_{L^\infty}$. The regularity gap closes only at second order, through the H-metric contraction: the viscous dissipation drives the attractor distance $H(u \mid \mathcal{A})$ towards a contradiction with its non-negativity. This three-level hierarchy is discussed in the broader context of the FST programme in [21].

Remark 3.41 (Spectral structure of the minimiser instability). The unstable directions of the distance functional $E_v = \frac{1}{2}\|u - v\|^2$ at the minimiser $v(t) \in \mathcal{A}$—those along which $\frac{d}{dt} E_v > 0$—correspond to the Lyapunov-unstable modes of the semiflow at $v(t)$, forming a finite-dimensional subspace of dimension at most $\dim(\mathcal{A})$. The time derivative $\frac{d}{dt} E_v$

decomposes into a dissipative part (controlled by $-\nu\|\nabla u\|^2$) and a Gronwall part (controlled by $C_1E_v+C_2$). Contraction in the H-metric is thus a *second-order* phenomenon: it stabilises the trajectory via the viscous term after the first-order projection has localised the instability to a finite-dimensional residuum.

Remark 3.42 (Measurable selection). The existence of a measurable map $t\mapsto v(t)\in\mathcal{A}$ realising $H(u(t)\mid\mathcal{A})=\frac{1}{2}\|u(t)-v(t)\|_{L^2}^2$ is a standard consequence of the Kuratowski–Ryll-Nardzewski measurable selection theorem applied to the compact-valued, lower-semicontinuous multifunction $t\mapsto\arg\min_{v\in\mathcal{A}}\|u(t)-v\|_{L^2}$; see e.g. [9], Theorem 4.1.

Remark 3.43 (Explicit dependence of constants). The constants appearing in the proof depend on the problem data as follows.

- C_{BS}: depends on $s>3/2$ and the geometry of $\mathbb{T}^3$ only (Lemma 3.1).
- $C_1=2\sup_{v\in\mathcal{A}}\|\nabla v\|_{L^\infty}+1$: finite because $\mathcal{A}$ is a compact attractor absorbing all orbits, so $\sup_{v\in\mathcal{A}}\|v\|_{W^{1,\infty}}<\infty$.
- Under Assumption G (AC version): $C_2=\frac{1}{2}\sup_{v\in\mathcal{A}}\|(v\cdot\nabla)v\|_{L^2}^2+\frac{1}{2}\|f-\partial_t v\|_{L^2}^2+\frac{\nu}{2}\sup_{v\in\mathcal{A}}\|\nabla v\|_{L^2}^2$. Under the weaker Assumption G′ (BV version), the minimiser velocity $\partial_t v$ is a measure rather than a function, and the corresponding term is replaced by the Stieltjes bound (28); the modified constant $\widetilde{C}_2=\frac{1}{2}\sup_{v\in\mathcal{A}}\|(v\cdot\nabla)v\|_{L^2}^2+\frac{1}{2}\|f\|_{L^2}^2+\frac{\nu}{2}\sup_{v\in\mathcal{A}}\|\nabla v\|_{L^2}^2$ does not contain $\|\partial_t v\|$. Both versions are finite by attractor compactness and the assumption $f\in L_t^\infty H_x^1$. In particular, both C_1 and C_2 (resp. $\widetilde{C}_2$) depend on the forcing amplitude $\|f\|_{L^\infty H^1}$, the viscosity ν, and the attractor radius (itself determined by ν and $\|f\|$), but are independent of T and u_0.

4 Discussion

4.1 Relation to Partial Regularity Results

The Caffarelli–Kohn–Nirenberg theorem [3] establishes that the one-dimensional parabolic Hausdorff measure of the singular set of any suitable weak solution is zero. While this is a profound geometric constraint, it does not exclude finite-time blow-up: it merely limits the size of the potential singular set. Similarly, the Serrin-type regularity criteria [7, 4] provide sufficient conditions for smoothness (e.g., $u\in L_t^pL_x^q$ with $2/p+3/q\le 1$), but verifying these conditions a priori requires information that is not available from the initial data alone.

Our approach differs structurally from both of these paradigms. Rather than imposing external integrability conditions on the solution or estimating the size of potential singularities, we exploit the internal dissipative structure of the equations: any blow-up scenario forces the *pointwise* enstrophy $\|\nabla u(t)\|_{L^2}^2\to\infty$ (via the BKM criterion and the enstrophy balance), creating a tension with the non-negativity of the attractor distance functional $H(u\mid\mathcal{A})$. Under the additional dissipation-dominance condition (D)—requiring that the cumulative viscous dissipation outpaces the Gronwall cost—this tension becomes a contradiction: $H<0$. The argument operates within the energy-dissipation framework, with two key remaining conditions: a regularity assumption on the attractor projection (Assumption G) and the dissipation-dominance condition (D).

4.2 Structural Advantages of the Attractor-Based Approach

Classical energy methods for the Navier–Stokes equations (see e.g. [4, 8]) typically produce Gronwall-type inequalities of the form

$$\frac{d}{dt}\|u\|_{H^1}^2 \le C\,\|u\|_{H^1}^2\,\|\nabla u\|_{L^\infty} - \nu\|\nabla u\|_{H^1}^2,$$

where the vortex-stretching term on the right-hand side competes with viscous dissipation. The fundamental difficulty is that no a priori bound on $\|\nabla u\|_{L^\infty}$ is available in three dimensions, and the Gronwall estimate merely yields a double-exponential bound that may blow up in finite time.

The attractor-based approach avoids this impasse by shifting the reference frame: instead of estimating $\|u\|_{H^1}$ directly, one measures the *deviation* $w = u - v$ from a smooth reference state $v \in \mathcal{A}$. The Gronwall coefficient $C_1 = 2\|\nabla v\|_{L^\infty} + 1$ is now *bounded* (since v lies on the compact attractor), replacing the uncontrolled $\|\nabla u\|_{L^\infty}$ of the classical approach. The price is Assumption G: one must control the time regularity of the attractor projection. This is a genuinely different problem from controlling $\|\nabla u\|_{L^\infty}$—it is a geometric property of the attractor rather than a pointwise bound on the solution.

5 Navier-Stokes Regularity as a Pattern A Stability System

Disclaimer. Sections 5.3–6 and the following remarks place the present framework in the context of the broader FST programme. These connections are *speculative and programmatic*: they sketch analogies and potential transfer paths between different problems, but none of them is mathematically substantiated in the present paper. The reader interested only in the self-contained conditional framework may skip directly to Section 6 (the H^1-lift programme).

The exclusion of finite-time blow-up in the 3D Navier-Stokes equations is framed here as a realization of the **Pattern A Stability Logic** (see [21]). This approach replaces the search for global a priori bounds on $\|\nabla u\|_{L^\infty}$ with the following structural architecture:

a) **Analytical Rank-Finiteness (Null-Tail):** Any hypothetical blow-up scenarios are localized to a finite-dimensional manifold of instability modes. The high-frequency (UV) dissipation scale is strictly controlled by the enstrophy balance [7], ensuring an analytical Null-Tail.

b) **Attractor-Projection as Physical Gauge:** The "finite negativity" (potential singularity directions) is neutralized by projecting the active flow onto the global Leray–Hopf attractor $\mathcal{A}$. This projection acts as the physical gauge.

c) **Constrained Positivity (H-Functional):** The non-negativity of the distance functional $H(u \mid \mathcal{A})$ acts as a gauge-stabilizer, ensuring that no trajectory can pay the "dissipation cost" of a blow-up without violating the thermodynamic budget.

Thus, global regularity is the manifestation of spectral stability under the gauge-constraint of the attractor structure. The dissipation integral acts as an irreversible "cost" that any blow-up trajectory must pay, while the attractor distance provides a non-negative "budget" that cannot be exceeded. This mechanism is analogous to the role of entropy production in non-equilibrium thermodynamics: the system's own dissipative structure forbids configurations requiring infinite entropy production in finite time.

5.1 Limitations and Open Questions

Several aspects of the present work invite further investigation:

- **Assumption G (the Gronwall gap).** The most significant limitation of this work is the reliance on Assumption G. The differential inequality for $H(u \mid \mathcal{A})$ in the proof of Theorem 3.35 requires that the minimising attractor element $v(t)$ evolves with sufficient time regularity. While every fixed $v \in \mathcal{A}$ is smooth in space, the *trajectory* of nearest-point projections $t \mapsto v(t)$ may exhibit jumps if the attractor has complicated geometry (e.g., cusps or self-intersections in the L^2-topology). Three strategies for closing this gap appear most promising: (i) proving the existence of an inertial manifold for 3D Navier–Stokes on $\mathbb{T}^3$, which would make the projection Lipschitz; (ii) establishing log-Lipschitz regularity of the projection via squeezing properties of the semiflow (cf. Barbu and Cannone); (iii) exploiting zero Lipschitz deviation on the attractor to show that high-frequency modes are deterministically enslaved by low-frequency modes, rendering $v(t)$ automatically smooth. Independent support for the attractor-geometric approach comes from recent work on fractal sparseness conditions: Grujic and Xu [22] show that if the regions of intense vorticity are sufficiently sparse (in a fractal-measure sense), singularities are topologically excluded—the first scaling reduction of the NS supercriticality gap since the 1960s. Their sparseness condition is structurally compatible with the finite-dimensional attractor assumed here.

- **Domain geometry.** Our proof relies on the periodic setting $\Omega = \mathbb{T}^3$, which ensures both the existence of a compact global attractor [5] and the vanishing of boundary terms in integration by parts. Extension to bounded domains with no-slip boundary conditions introduces boundary-layer effects and requires attractor theory in $H^1_0(\Omega)$, which is available (cf. [8]) but technically more involved. The whole-space case $\Omega = \mathbb{R}^3$ is more delicate, as the global attractor may not be compact in the usual sense; one would need to work with trajectory attractors or pullback attractors instead.

- **External forcing.** We have assumed smooth, time-independent (or at least uniformly bounded) forcing $f \in L^\infty_t H^1_x$. For rough or singular forcing, the uniform bounds on C_1 and C_2 may fail, and the attractor structure itself can change. Extending the approach to stochastic forcing (as in the stochastic Navier–Stokes equations) is an interesting direction that would require a random attractor framework.

- **Quantitative bounds.** The present framework is conditional: it does not prove global regularity unconditionally, but rather reduces the problem to Assumptions G and (D). Should both conditions be verified in the future, the resulting bounds on $\|\nabla u\|_{L^\infty}$ would be non-constructive, depending on the attractor radius and the absorbing ball estimates of the Navier–Stokes semiflow. Deriving explicit, quantitative regularity bounds from the attractor distance would require more precise information about the attractor geometry, which remains largely unknown in three dimensions.

- **Measurable selection regularity.** The proof uses a measurable selection $t \mapsto v(t) \in \mathcal{A}$ to track the nearest attractor element. While measurability suffices for the integral estimates, higher regularity of this selection (e.g., absolute continuity) would strengthen the differential inequality and potentially yield sharper decay rates for $H(u \mid \mathcal{A})$.

Remark 5.1 (Breaking the regularity–attractor circularity). The standard approach to global attractors in dissipative PDE theory (Temam [8], Zelik [17]) assumes smoothness of solutions to prove finite-dimensionality of the attractor. Our framework inverts this logical direction: we assume finite-dimensionality of the attractor (which is empirically well-supported and established for the weak Leray–Hopf attractor; cf. Remark 2.3) and derive regularity consequences.

The apparent circularity—regularity is needed for the attractor, and the attractor is needed for regularity—is broken by the *thermodynamic obstruction*: if finite-time blow-up occurred, the free energy functional $H(u \mid \mathcal{A})$ would diverge to $-\infty$, contradicting the dissipative bound $H(u \mid \mathcal{A}) \geq 0$. The non-negativity of H is a consequence of the definition (H is a squared distance), not of regularity.

This is *not* a proof of unconditional regularity, but a *structural reduction*: global regularity follows IF the attractor has the assumed properties (finite dimensionality, compactness, smoothness of elements). Each of these properties is established independently of the regularity question—they follow from the weak attractor theory and the bootstrap argument for elements on the attractor (Proposition 2.2). The remaining gap is Assumption G, which concerns the *time regularity* of the projection, not the spatial regularity of the attractor elements.

Remark 5.2 (Obstructions to obstruction arguments and non-uniqueness). Two results from the recent literature deserve explicit mention as structural warnings for the present framework.

(i) Tao's averaged blow-up. Tao [51] proved that a certain averaged version of the 3D Navier–Stokes equations—which preserves the energy identity, the enstrophy structure, and many of the PDE's symmetries—exhibits finite-time blow-up. This demonstrates that *no obstruction argument that relies solely on energy-level quantities and general dissipative structure can succeed*: additional information specific to the genuine Navier–Stokes nonlinearity (as opposed to its averaged surrogate) is required. In the present framework, this additional information is encoded in Assumption G, which concerns the geometry of the *actual* NS attractor (not a generic dissipative attractor).

(ii) Non-uniqueness of Leray–Hopf solutions. Buckmaster and Vicol [52] constructed non-unique weak solutions satisfying the energy inequality, calling into question whether the Leray–Hopf semiflow is well-defined as a single-valued map. For the attractor-based approach, this means that the attractor $\mathcal{A}$ may depend on the choice of selection among Leray–Hopf solutions (see also the selection dependence discussed in Remark 2.3). Theorem 3.35 holds for any fixed selection; the non-uniqueness does not invalidate the conditional framework but underscores that the attractor is defined relative to a chosen semiflow.

Remark 5.3 (Decomposition of Assumption G: independent components). To address the concern that Assumption G might be circular (implicitly assuming regularity), we decompose it into three logically independent components:

(G$_1$) Attractor existence and compactness. The global attractor $\mathcal{A}$ exists as a compact subset of $H^1(\Omega)$ and is finite-dimensional ($d_H(\mathcal{A}) < \infty$). *Status:* Established for the weak Leray–Hopf attractor without assuming unconditional regularity (Foias–Temam, Sell, Chepyzhov–Vishik; cf. Remark 2.3). This component is **independent** of the regularity question.

(G$_2$) **Projection regularity.** The nearest-point projection $\pi_{\mathcal{A}} : H^1(\Omega) \to \mathcal{A}$ is log-Lipschitz (or admits a BV selection) in a tubular neighbourhood of $\mathcal{A}$. *Status:* Open. This is a **geometric** property of the attractor, not a dynamical regularity statement. It can in principle be verified from the attractor's metric structure (Hausdorff dimension, tangent cone regularity, Assouad dimension) without reference to the regularity of solutions.

(G$_3$) **Gronwall integrability.** The time-dependent Gronwall coefficients $C_1(t), C_2(t)$ in the differential inequality for $H(u|\mathcal{A})$ satisfy $\int_0^T C_i(t)\,dt < \infty$. *Status:* Follows from standard energy estimates and the absorbing ball property, assuming the solution remains in H^1. Under blow-up, the enstrophy $\|\nabla u\|^2$ diverges, making C_1 unbounded—but this is precisely the contradiction mechanism of Theorem 3.35.

The circularity concern applies only to (G$_2$): if one proves regularity first, then the attractor is smooth and (G$_2$) is trivial. Our framework breaks this circularity by treating (G$_2$) as a *geometric hypothesis* about the attractor (verifiable from its metric properties) rather than a dynamical regularity assumption. The bridge from (G$_1$) to (G$_2$) is the content of the companion paper on Log-Distance Integrability [41].

Remark 5.4 (Assumption G as interface contract). Assumption G can be decomposed into a minimal *interface contract* specifying the exact regularity data required from the attractor geometry:

Input 1 (Attractor regularity): $\sup_{v\in\mathcal{A}} \|v\|_{H^2} < \infty$. *Status: known* (follows from smoothness of the global attractor).

Input 2 (Selection regularity): The nearest-point map $t \mapsto v(t) \in \mathcal{A}$ satisfies $v \in BV([0,T]; H^1)$. *Status: open*—this is the core gap, addressed by the TLL/LDI framework of the companion paper [41].

Input 3 (Gronwall integrability): The Gronwall coefficients $C_1(t)$ satisfy $\int_0^{T^*} C_1(t)\,dt < \infty$. *Status: follows from Input 2* (BV selection implies locally integrable derivative bounds).

This decomposition clarifies that the entire open content of Assumption G reduces to a single question: *does the nearest-point projection onto the attractor have bounded variation along Leray–Hopf solutions?* The companion paper provides a sufficient condition (TLL + LDI) and validates it numerically on three attractors.

Remark 5.5 (Failure modes of Assumption G). If Assumption G fails, the failure has specific geometric signatures:

1. *Infinite jump accumulation:* The minimiser $v(t) = \arg\min_{v\in\mathcal{A}} \|u(t) - v\|^2$ experiences infinitely many discontinuous jumps in any neighbourhood of T^*, with jump sizes not summable. Geometrically, this corresponds to the trajectory $u(t)$ crossing infinitely many "ridges" of the Voronoi tessellation of $\mathcal{A}$ in finite time.

2. *Cantor-type switching:* The set of switching times $\{t : v(t^-) \neq v(t^+)\}$ has positive measure or Cantor-like structure, preventing BV regularity. This would require the attractor to have a pathologically fragmented Voronoi structure at the relevant scales.

3. *Spiral accumulation:* The trajectory spirals around a cusp or self-intersection of $\mathcal{A}$, causing the projection to oscillate with increasing frequency. This is the most physically plausible failure mode and would indicate that the attractor geometry near potential blow-up points is more complex than current dimension estimates suggest.

The TLL/LDI framework of the companion paper [41] guards against all three modes: TLL bounds the oscillation rate logarithmically, while LDI ensures the cumulative oscillation cost remains finite.

Remark 5.6 (Reach lemma via backward stability). If the attractor A has positive reach $\mathrm{reach}(A) > 0$ (i.e., the nearest-point projection is single-valued in a tubular neighbourhood), then backward stability (solutions on A are Lipschitz in backward time) combined with injectivity of finite-dimensional projections $P_N : A \to \mathbb{R}^N$ implies $\mathrm{reach}(A) \geq c/\|P_N\|_{\mathrm{op}}$. This would establish Assumption G (Lipschitz regularity) as a consequence of finite-dimensionality, breaking the circularity noted in the limitations.

Remark 5.7 (Relaxation of Condition (D) to limsup divergence). Condition (D) can be relaxed from pointwise dominance to a measure-theoretic version: instead of requiring $\|\nabla u(t)\|^2/H(u(t)|A) \to \infty$ as $t \to T^*$ for all blow-up trajectories, it suffices to demand

$$\limsup_{t \to T^*} \frac{\|\nabla u(t)\|^2}{H(u(t)|A)} = +\infty.$$

This weaker condition still forces the contradiction in Theorem 3.1 (the free energy barrier cannot be sustained if dissipation intermittently dominates), and is more natural from the PDE perspective: enstrophy blow-up need not be monotone, only unbounded.

Remark 5.8 (Condition D: observable checklist). The following observable quantities in DNS or numerical simulation would provide evidence for or against Condition D:

1. *Enstrophy growth rate:* If $\|\nabla u(t)\|^2_{L^2}/H(u(t)|\mathcal{A})$ grows without bound as $t \to T^*$, Condition D holds. Monitor the ratio $\|\nabla u\|^2/\|u - v\|^2$ along near-singular trajectories.

2. *Palinstrophy:* The H^1-level analogue requires $\int_0^{T^*} \|\Delta u\|^2_{L^2}\, dt = \infty$. Track the cumulative palinstrophy in adaptive-resolution simulations approaching potential blow-up.

3. *Attractor distance decay:* If $H(u(t)|\mathcal{A}) \to 0$ faster than $\|\nabla u\|^{-2}$, Condition D is satisfied. Compare the rates numerically.

A falsification of Condition D (enstrophy ratio remaining bounded through blow-up) would indicate that the attractor-distance approach requires modification, but would not resolve regularity—it would merely shift the problem to finding a different Lyapunov-type functional.

5.2 A Game-Theoretic Perspective

The regularity question admits a natural reformulation as a minimax problem. Consider two abstract "players" governing the dynamics: Player D (Dissipation), whose strategy is viscous damping at rate $\nu\|\nabla u\|_{L^2}^2$, and Player N (Nonlinearity), whose strategy is the convective energy transfer via $(u \cdot \nabla)u$. Define the *game value* at mode level as

$$\mathcal{V}(t) \ := \ \inf_{\text{modes}} \Big[\text{dissipation rate} - \text{nonlinear transfer rate}\Big]. \tag{30}$$

In this language, global regularity is equivalent to $\mathcal{V}(t) \geq 0$ for all t: dissipation dominates the nonlinearity uniformly across all Fourier modes. The attractor distance $H(u \mid \mathcal{A})$ plays the role of a finite "budget" available to the system, while the dissipation integral $\int_0^T \nu\|\nabla u\|_{L^2}^2\, dt$ represents the cumulative cost that any blow-up trajectory must pay. Since the budget is non-negative, the question reduces to whether the cumulative dissipation cost exceeds the Gronwall cost near the blow-up time—precisely the dissipation-dominance condition (D) of Theorem 3.35.

This viewpoint connects to the Legendre–Fenchel duality in convex analysis: interpreting the dissipation as a convex "entropy production" functional S and the nonlinear transfer as its conjugate "free energy cost" F, the regularity condition $S \geq F$ is the PDE analogue of the second law of thermodynamics—the system cannot extract more work from the nonlinearity than the dissipation permits.

Remark 5.9 (Mean field game perspective). The Fokker–Planck component of Mean Field Game systems (Lasry–Lions [6]) has the form $\partial_t m - \nu\Delta m + \nabla \cdot (m\alpha) = 0$, structurally identical to advection-diffusion with drift α. A Nash equilibrium—where infinitely many "fluid particles" optimise against the collective density—corresponds to a self-consistent balance of transport and diffusion. Whether the well-posedness theory of MFG systems can inform the Navier–Stokes regularity question is an open direction.

Remark 5.10 (Mean field game formulation in Wasserstein space). Recent developments in mean field game (MFG) theory formulate the coupled HJB–Fokker-Planck system as a PDE on the Wasserstein space of probability measures (Cardaliaguet–Delarue–Lasry–Lions [49]). The NS free energy $H(u|A)$ can be interpreted as a value function in this MFG framework: the "agent" (fluid parcel) minimises a running cost (enstrophy) subject to advection-diffusion dynamics, while the "population" (velocity field) evolves via the Navier–Stokes equations. The attractor A corresponds to the MFG equilibrium. This perspective suggests that regularity of the NS attractor is equivalent to well-posedness of the associated master equation in Wasserstein space.

Remark 5.11 (Attractor–Gronwall control and Nash frustration). The Attractor–Gronwall Control (AGC) condition of Section 3 admits an instructive parallel to the *Nash frustration* concept developed in FST-III [21]. In game-theoretic terms, Nash frustration arises when no pure Nash equilibrium exists, forcing the system into a mixed-strategy regime. The Navier–Stokes attractor $\mathcal{A}$ plays a structurally analogous role: when smooth individual solutions ("pure strategies") fail to persist globally—i.e., when the Gronwall budget is exhausted—the dynamics is forced onto the attractor, which acts as the "mixed strategy" that absorbs the frustrated energy budget. The AGC condition quantifies the cost of this transition: if the attractor projection $t \mapsto v(t)$ is sufficiently regular (positive reach or BV-selection), the system can redistribute the dissipation cost across the attractor manifold without producing a blow-up singularity. In this reading, global regularity is the statement that the Navier–Stokes system always finds a "mixed strategy" (the attractor) to resolve the Nash frustration of competing vortex-stretching and viscous-damping modes.

Remark 5.12 (Concentration of the attractor-distance functional)*.* In the dissipative regime, the enstrophy balance (Lemma 3.3) prevents large excursions of H and supports concentration of trajectories around $\mathcal{A}$. If H admits sub-Gaussian exponential moments along the trajectory, all higher cumulants are variance-dominated, providing quantitative tail bounds for $\{H > \ell\}$. This does not affect the deterministic argument of Theorem 3.35 but motivates probabilistic extensions; see Kuksin–Shirikyan [10] and Hairer–Mattingly [11] for related invariant-measure approaches.

Remark 5.13 (Pattern A Master Stability – cross-problem structure)*.* This paper instantiates the Pattern A Master Stability Principle from the unified framework [21]: strict convexity of the renormalised free energy F, combined with a neutral leading resolvent branch and second-order dominance, implies a spectral gap $\Delta > 0$ for the associated transfer operator. The specific instantiation in the present context is: $\Delta =$ exponential relaxation rate to the global attractor.

Remark 5.14 (Wasserstein convexity and Talagrand bound)*.* The free energy functional $H(u|A)$ is uniformly convex in the Wasserstein-2 metric with convexity constant $\lambda_H > 0$ (inherited from the quadratic structure of the H^1-norm). By the Otto–Villani theorem, this implies a Talagrand inequality $W_2(\mu, \mu^*) \leq \sqrt{2D_{\mathrm{KL}}(\mu\|\mu^*)/\lambda_H}$, which provides a quantitative bound on how far any trajectory distribution can deviate from the attractor measure. The Poincaré constant satisfies $\kappa \geq \lambda_H$, giving a β-independent lower bound—in contrast to the Holley–Stroock bound, which degrades exponentially. This structural observation, first formalised in the Yang–Mills companion paper, applies universally to all FST instantiations.

5.3 Transfer from Yang–Mills: Poincaré–Dobrushin on Galerkin-truncated Navier–Stokes

The Galerkin truncation of the Navier–Stokes equations at mode number N is structurally analogous to the lattice Yang–Mills regularisation at lattice spacing $a \sim 1/N$:

	Yang–Mills (lattice)	**Navier–Stokes (Galerkin)**
Degrees of freedom	Link variables $U_l \in G$	Fourier modes $\hat{u}_k \in \mathbb{C}^3$
Interaction range	Plaquettes (nearest-neighbour)	Triadic interactions $k + p + q = 0$
Regularisation	Lattice spacing a	Mode cutoff N
Continuum limit	$a \to 0$ $(\beta \to \infty)$	$N \to \infty$
Free energy	$F[\mu] = \langle S \rangle + D_{\mathrm{KL}}$	$H(u|A) = \frac{1}{2}\|u - u^*\|^2_{H^1}$

Remark 5.15 (Dobrushin influence matrix for Galerkin modes)*.* Define the Dobrushin interdependence matrix $C = (c_{kp})$ for Galerkin modes by

$$c_{kp} = \sup \|\mu_k(\cdot|\omega) - \mu_k(\cdot|\omega')\|_{\mathrm{TV}},$$

where the supremum is over configurations differing only in mode p, and $\mu_k(\cdot|\omega)$ is the conditional distribution of mode k given all other modes. For the Navier–Stokes nonlinearity $B(u, u)$, the entry c_{kp} is non-zero only when there exists q with $k+p+q = 0$ (triadic interaction). The spectral radius satisfies

$$\rho(C) \leq \sup_k \sum_{p:\exists q, k+p+q=0} c_{kp} \leq C_{\mathrm{tri}} \cdot \frac{\|u\|_{H^1}}{N^{1/2}},$$

where C_{tri} is a universal constant and the $N^{-1/2}$ decay reflects the decreasing influence of high modes. The Dobrushin uniqueness condition $\rho(C) < 1$ holds for sufficiently large N (i.e., sufficiently fine Galerkin truncation), providing a Poincaré inequality for the truncated dynamics.

The key difference from Yang–Mills is that the NS interaction is *long-range* in Fourier space (all triadic interactions contribute), whereas YM is nearest-neighbour on the lattice. However, the energy cascade ensures that the effective interaction strength decays with scale separation $|k - p|$, recovering a quasi-local structure in the inertial range.

Remark 5.16 (Gap transport analogue for Galerkin truncations). The analogue of the Yang–Mills gap transport conjecture in the NS context is: the Poincaré constant of the Galerkin-truncated system at cutoff N satisfies $\kappa(2N) \geq c \cdot \kappa(N)$ for a universal $c > 0$. If this holds, the attractor's exponential relaxation rate persists in the limit $N \to \infty$, providing an independent route to regularity.

5.4 Open directions

Remark 5.17 (Inertial manifold path to Assumption G). The theory of inertial manifolds [23] provides the most direct path to closing Assumption G. On an inertial manifold $\mathcal{M}$, the high-frequency modes $Q_N u$ are deterministically enslaved to the low-frequency modes $P_N u$ via a Lipschitz map $\Phi\colon P_N H \to Q_N H$, so that the nearest-point projection $t \mapsto v(t)$ inherits Lipschitz regularity from the finite-dimensional flow on $\mathcal{M}$. The zero Lipschitz deviation $\mathrm{dev}_m(\mathcal{A}) = 0$ [14] would then imply that Assumption G holds automatically: the AC regularity of $v(t)$ follows from the finite-dimensional ODE on the manifold.

In the companion turbulence paper [19], the same machinery is used to formalise the K41 spectrum as the saddle point of a scale-specific KL-divergence on the inertial manifold. The structural parallel is: inertial manifolds reduce both the NS regularity problem and the turbulence selection problem to finite-dimensional variational questions where the KL framework is rigorous.

The main obstruction is the *spectral gap condition*: inertial manifolds for 3D Navier–Stokes on $\mathbb{T}^3$ require a gap between consecutive eigenvalues of the Stokes operator that is not known to hold for $\beta = 1$ (standard viscosity). For hyperviscous NS ($\beta > 5/4$), inertial manifolds exist [18]; extending this to $\beta = 1$ remains a major open problem [17]. Recent work on the fractional Navier–Stokes–Voigt system [15] provides sharp attractor-dimension estimates via Lieb–Thirring inequalities that improve the classical non-fractional bounds and may guide analogous estimates in the present framework.

Remark 5.18 (Log-Dirichlet quotients in the KL joint minimisation). An alternative to the inertial manifold approach is to integrate *log-Dirichlet quotients* for differences of solutions on the attractor into the KL-joint-minimisation framework from the turbulence companion paper [19]. For two solutions u, v on the attractor, define the log-Dirichlet quotient

$$Q_{\mathrm{LD}}(t) = \frac{\|\nabla(u - v)\|_{L^2}^2}{\|u - v\|_{L^2}^2}.$$

If Q_{LD} is integrable along trajectories, the Gronwall estimate for $H(u \mid \mathcal{A})$ transforms from exponential to algebraic growth: the Lyapunov exponents are stabilised, and the cumulative Gronwall cost $\int_0^{T^*} Q_{\mathrm{LD}}(s)\,ds$ remains finite even as $T^* \to T^*_{\max}$.

This algebraic (rather than exponential) growth topologically forbids finite-time singularities: the free energy $H(u \mid \mathcal{A})$ can decrease at most polynomially, which is incompatible with the divergent enstrophy required for blow-up.

Remark 5.19 (Helicity stratification of the attractor). The global attractor $\mathcal{A}$ can be decomposed into helicity layers $\mathcal{A} = \bigcup_h \mathcal{A}_h$, where $\mathcal{A}_h = \{u \in \mathcal{A} : \mathcal{H}(u) = h\}$ and $\mathcal{H}(u) = \int \mathbf{u} \cdot \boldsymbol{\omega}\, dx$ is the helicity [24, 25]. Within each layer $\mathcal{A}_h$, the topology constrains the vortex-line geometry, stabilising the nearest-point projection: the *stratified* projection $v_h(t) = \arg\min_{w \in \mathcal{A}_h} \|u(t) - w\|$ restricts the minimiser search to a topologically coherent subset, potentially improving the regularity of $t \mapsto v_h(t)$ and thereby grounding Assumption G on controlled, topology-respecting selection rather than naked arg min over the full attractor.

Remark 5.20 (Energy-helicity inequalities as topological coercivity). For knotted vortex configurations, topology enforces a lower bound on the gradient energy [25]:

$$\|\nabla u\|_{L^2}^2 \geq c_0\, |\mathcal{H}(u)|^{3/4},$$

where $c_0 > 0$ depends on the domain and the Faddeev–Skyrme-type energy bound. This topological coercivity supplements the $H(u \mid \mathcal{A})$ budget with a lower bound that is purely geometric: even without solving the Gronwall inequality, the topology of vortex lines constrains how fast the enstrophy can grow, potentially providing an independent obstruction to blow-up.

Remark 5.21 (Knot complexity as attractor regulariser). Define a functional knot complexity $K(u)$ on the attractor via the helicity density spectrum and linking statistics of the vortex lines [24]. If K is uniformly bounded on $\mathcal{A}$—which is plausible given the finite-dimensionality and compactness of the attractor—then the minimiser variation $\|v(t_1) - v(t_2)\|$ is controlled by the topological distance between the corresponding knot classes, providing a geometric mechanism for the time-regularity of the projection $t \mapsto v(t)$.

Remark 5.22 (Helicity cascade as downhill motion in knot space). The helicity stratification of the attractor (Remark 5.19) has a direct counterpart in the turbulence companion paper [19]: the DFC remark in the turbulence companion paper argues that vortex reconnections are dissipative steps that lower the topological complexity of the vorticity field, making the energy cascade "downhill" in knot space. In the NS regularity context, the same mechanism stabilises the attractor projection: transitions between helicity layers $\mathcal{A}_h \to \mathcal{A}_{h'}$ are dissipative (helicity is nearly conserved and decreases via reconnection), so the projection $t \mapsto v_h(t)$ within a given layer evolves more smoothly than an unrestricted arg min. This bridge—cascade topology (TU) $\leftrightarrow$ attractor topology (NS)—suggests that any resolution of the Downhill Flux Condition in turbulence simultaneously constrains the attractor geometry relevant for Assumption G.

Remark 5.23 (Grujic–Xu sparseness as Dobrushin-type uniqueness). The fractal sparseness condition of Grujic and Xu [22]—regions of intense vorticity occupy a set of fractal dimension strictly less than 5/3 in space-time—is structurally analogous to the Dobrushin uniqueness condition exploited in the Yang–Mills companion paper and in Remark 5.15: both assert that *interactions are sufficiently sparse* to ensure effective locality. In YM, the Dobrushin influence matrix C satisfies $\rho(C) < 1$ (exponential decay of correlations); in NS, the sparseness condition ensures that vortex stretching events are too thin to accumulate into a singularity. The common principle is: *geometric sparseness* $\Rightarrow$ *effective locality* $\Rightarrow$ *uniqueness/regularity.*

Remark 5.24 (Finite-dimensional attractor implies vorticity sparseness). The finite fractal dimension $d_f(\mathcal{A}) < \infty$ of the global attractor implies that on-attractor solutions are confined to a "thin" subset of phase space. If this thinness propagates to the *physical-space* vorticity distribution—i.e., if solutions on $\mathcal{A}$ satisfy the Grujic–Xu sparseness condition automatically—then the fractal-dimension bound provides an *independent route to Assumption G* that bypasses the spectral gap barrier of inertial manifolds entirely. The key conjecture is:

$$d_f(\mathcal{A}) \le D_0 \implies \dim_{\mathcal{H}}(\{x : |\omega(x)| > \lambda\}) \le 3 - c(\lambda, D_0) \quad \text{for all } \lambda \gg 1,$$

where $c > 0$ grows with λ, yielding Grujic–Xu-type sparseness. Establishing this implication would close Assumption G for attractors of known finite dimension, which includes the 2D case and conjectured 3D scenarios.

Remark 5.25 (Advanced approaches to attractor regularity). We collect four speculative but concrete approaches to establishing attractor regularity:

1. *Multiplicative Oseledets cones:* Invariant convexity cones in the tangent space $T_u A$ at attractor points, inherited from the monotone structure of the Navier–Stokes semigroup, could provide cusp-smoothing: if the cone angle is uniformly bounded, cusps (and hence singularities) are excluded.

2. *Fractional embedding $H^{1+\varepsilon}$:* Log-Lipschitz regularity of the attractor projection (established in the companion LogDist paper) may suffice for smooth normal bundles, bypassing the full Lipschitz requirement of Assumption G.

3. *Dissipation as Radon measure:* Reinterpreting the energy dissipation $\varepsilon(t) = \nu\|\nabla u\|^2$ as a Radon measure on $[0, T^*) \times \mathbb{R}^3$, Condition (D) becomes strict positivity of the singular part $\varepsilon_{\text{sing}}$—a measure-theoretic condition that connects to the Caffarelli–Kohn–Nirenberg partial regularity theory.

4. *Compensated compactness:* Tartar's compensated compactness method applied to vorticity oscillations ($\omega = \nabla \times u$) may force integral divergence of $\|\nabla u\|^2$ near a hypothetical singularity, establishing Condition (D) from the structural properties of the Euler nonlinearity alone.

Remark 5.26 (Spectral gaps survive conical singularities (Sturm)). Sturm [30] has recently shown that Bakry–Émery curvature conditions and spectral gap estimates remain valid on Riemannian manifolds with conical singularities, even when the Ricci curvature degrades as $\text{Ric} \sim 1/d^2$ near the singular points. Crucially, generalised Hardy inequalities replace the missing curvature-dimension conditions, and the spectral gap exhibits remarkable robustness against local topological defects.

This result has direct implications for the FST approach to Navier–Stokes regularity: even if the global attractor A has $\text{reach}(A) = 0$ (fractal boundary with cusp-like singularities), the thermodynamic relaxation mechanism — controlled by the spectral gap of the associated transfer operator — remains intact, provided the singularities are of conical type. This supports the weaker condition of time-averaged reach $\text{reach}_{\text{avg}}(A) > 0$ (cf. Remark 5.6): the Moreau–Yosida regularised selection "sees" only the average geometry, which is smooth even if pointwise cusps exist.

Remark 5.27 (Duchon–Robert defect measure as bridge between turbulence and attractor regularity). Duchon and Robert [29] introduced a local energy defect measure $D(u)$ for weak solutions of the incompressible Euler and Navier–Stokes equations. For a weak solution u, the distribution $D(u)$ captures the local rate of inertial energy dissipation beyond the viscous scale: it is the distributional remainder in the local energy balance

$$\partial_t \tfrac{|u|^2}{2} + \nabla \cdot \Big[\Big(\tfrac{|u|^2}{2} + p\Big)u\Big] + \nu\|\nabla u\|^2 = -D(u).$$

A key physical constraint is $D(u) \geq 0$ almost everywhere for physically admissible solutions (energy flows only to smaller scales).

This defect measure connects directly to the FST attractor framework in three ways:

1. *Singularity localisation on the attractor.* On the global attractor $\mathcal{A}$, the support of $D(u)$ encodes the singularity structure: blow-up can occur only where $D(u)$ concentrates. If the singular support of $D(u)$ has Hausdorff dimension $\dim_H(\operatorname{supp}_{\text{sing}} D) < 1$, then the BV selection $t \mapsto v(t)$ along the attractor is "almost" Lipschitz in the sense that its total variation is controlled by the $(1 - \dim_H)$-Hölder norm of the distance function.

2. *Coupling to BV selection theory.* The balanced-viscosity selection from the LDI paper [41] produces a trajectory $v \in \mathrm{BV}$ that minimises a dissipation functional. The defect measure $D(u)$ provides a *natural weight* for this minimisation: replacing the uniform L^2-distance by the D-weighted distance $\int D(u(t))\, \|u(t) - a(t)\|^2\, dt$ concentrates the selection where inertial dissipation is active, improving regularity in quiescent regions.

3. *Bridge to the turbulence companion paper.* In [19], the defect measure $D(u)$ plays the role of the anomalous dissipation flux in the turbulent cascade. The present paper's Assumption G concerns regularity of the projection onto $\mathcal{A}$; the turbulence paper concerns the spectral distribution of $D(u)$. A unified closing strategy would establish both by showing that $D(u)$ is a Radon measure with controlled Hausdorff-dimensional support—precisely the condition under which BV selection inherits near-Lipschitz regularity (cf. Remark 3.17).

5.5 EDP-convergence route to selection regularity

The classical pointwise projection $v(t) = \pi_A(u(t))$ is structurally inadequate: BV regularity is a time-integrated property, but pointwise projection ignores temporal coherence. We replace it by a *trajectory selection* that derives BV regularity from dissipation alone, without requiring positive reach, inertial manifolds, or spectral gaps.

Step 1: Rescaled trajectory functional

Let $\mathcal{A}_T$ denote a compact set of trajectories on or near the attractor (in the sense of Chepyzhov–Vishik trajectory attractors [37], which are available for 3D Navier–Stokes without uniqueness assumptions). For $\varepsilon > 0$, define the *rescaled* functional

$$\mathcal{F}_\varepsilon[a] := \frac{1}{\varepsilon}\int_0^T \|u(t) - a(t)\|_H^2\, dt \;+\; \mathrm{TV}(a), \tag{31}$$

where $\mathrm{TV}(a) = \sup \sum_i \|a(t_{i+1}) - a(t_i)\|_H$ is the total variation. Note the crucial rescaling: the fidelity term carries $1/\varepsilon$, while TV is unweighted. The regularised selection is

$$v_\varepsilon := \operatorname{argmin}_{a \in \mathcal{A}_T} \mathcal{F}_\varepsilon[a]. \tag{32}$$

Step 2: A priori bounds

For any minimiser v_ε, comparison with a fixed trajectory $a_0 \in \mathcal{A}_T$ gives

$$\mathrm{TV}(v_\varepsilon) \;\le\; \mathcal{F}_\varepsilon[v_\varepsilon] \;\le\; \mathcal{F}_\varepsilon[a_0] \;=\; \frac{1}{\varepsilon}\int_0^T \|u - a_0\|^2\,dt + \mathrm{TV}(a_0). \tag{33}$$

Remark on the a priori bound. For a *fixed* comparator $a_0 \in \mathcal{A}_T$, the right-hand side of (33) is $O(1/\varepsilon)$ (not $O(1)$), since $\int \|u - a_0\|^2 > 0$ in general. To obtain a *uniform* TV bound, one must choose an ε-dependent comparator $a_0^{(\varepsilon)} \in \mathcal{A}_T$ satisfying $\int_0^T \|u - a_0^{(\varepsilon)}\|^2\,dt = O(\varepsilon)$, i.e., the trajectory attractor must approximate u with accuracy $O(\sqrt{\varepsilon})$ in L^2. The existence of such approximations in $\mathcal{A}_T$ is itself a regularity assumption on u: it holds whenever u lies within the absorbing ball of the attractor and is bounded away from the singular set, but *may fail* in the blow-up regime. More precisely, the required $O(\varepsilon)$-approximability is a form of Assumption G (proximity of u to the trajectory attractor).

Under this additional assumption, the comparison gives:

$$\mathrm{TV}(v_\varepsilon) \;\le\; \mathcal{F}_\varepsilon[a_0^{(\varepsilon)}] \;=\; O(1) + \mathrm{TV}(a_0^{(\varepsilon)}) \;\le\; C \quad \text{uniformly in } \varepsilon. \tag{34}$$

This uniform bound is the **essential improvement** over the unrescaled formulation, where only $\varepsilon\,\mathrm{TV} \le C$ holds and TV blows up. However, the bound depends on the approximability assumption, which connects the EDP route back to Assumption G.

Step 3: Compactness via Simon's BV embedding

The uniform BV bound (34) alone does not yield compactness in $L^2(0,T;H)$ when H is infinite-dimensional (Helly's theorem fails). We invoke Simon's generalisation of the Aubin–Lions lemma to BV time regularity [38]:

Lemma 5.28 (Simon compactness for BV selections)**.** *Let $V \hookrightarrow\!\!\!\rightarrow H \hookrightarrow V'$ be a Gelfand triple with compact embedding $V \hookrightarrow H$. Suppose $\{v_\varepsilon\}_{\varepsilon>0}$ satisfies:*

(S1) $\|v_\varepsilon\|_{L^2(0,T;V)} \le C_1$ *(spatial regularity),*

(S2) $\mathrm{TV}_{V'}(v_\varepsilon) := \sup_{partitions} \sum_i \|v_\varepsilon(t_{i+1}) - v_\varepsilon(t_i)\|_{V'} \le C_2$ *(temporal BV in V').*

Then $\{v_\varepsilon\}$ is relatively compact in $L^2(0,T;H)$:

$$\{v_\varepsilon\} \textit{ is relatively compact in } L^2(0,T;H). \tag{35}$$

Proof sketch. By (S1), each $v_\varepsilon(t) \in V$ for a.e. t. By (S2), the time oscillations are controlled in V'. The compact embedding $V \hookrightarrow\!\!\!\rightarrow H$ and an interpolation argument (Simon [38], Theorem 5) yield equicontinuity in H, hence relative compactness via Arzelà–Ascoli in $L^2(0,T;H)$. □

Verification of (S1)–(S2) for trajectory-attractor selections.

- (S1): Trajectory-attractor elements $a(\cdot) \in \mathcal{A}_T$ satisfy the Navier–Stokes energy inequality, which gives uniform bounds in $L^2(0,T;V)$ with $V = H_0^1(\Omega)$.
- (S2): The BV bound (34) gives $\mathrm{TV}_H(v_\varepsilon) \le C$. Since $H \hookrightarrow V'$, this implies $\mathrm{TV}_{V'}(v_\varepsilon) \le C$.

After subsequence extraction: $v_\varepsilon \to v$ in $L^2(0,T;H)$, with $\mathrm{TV}(v) \le C$.

Step 4: EDP-convergence structure

The limit selection v satisfies:

1. $v(t) \in \overline{\mathcal{A}}$ a.e. (closedness of the trajectory attractor),

2. $\mathrm{TV}(v) \leq C \int_0^T \|\partial_t u\| \, dt$ (from the energy-dissipation balance of NS),

3. v minimises $\mathcal{F}_0[a] = \int_0^T \|u - a\|^2 \, dt$ over all BV selections on $\overline{\mathcal{A}}$.

Property (2) is *precisely* the BV selection bound required by Assumption G. It arises not from "dividing by ε" (which is invalid in the unrescaled setting) but from the Mielke–Rossi–Savaré EDP-convergence framework [32]: the energy-dissipation balance propagates through the Γ-limit via lower semicontinuity and chain-rule arguments, as in balanced-viscosity solutions [32].

Step 5: Balanced-viscosity formulation

The limit passage $\varepsilon \to 0$ in Step 4 can be made rigorous using the *balanced viscosity* (BV) framework of Mielke–Rossi–Savaré [32]. Rather than dividing by ε (which is invalid), the BV approach characterises the limit selection v as the solution of a *rate-independent variational inequality* coupled to jump costs:

Definition 5.29 (Balanced-viscosity selection)**.** A trajectory $v \in BV(0,T;\overline{\mathcal{A}})$ is a *balanced-viscosity selection* for u if it satisfies the energy-dissipation balance

$$\underbrace{\|u(t) - v(t)\|_H^2}_{\text{energy at } t} + \underbrace{\mathrm{Var}_{\mathrm{BV}}(v;[s,t])}_{\text{dissipation}} + \underbrace{\sum_{\tau \in J(v) \cap [s,t]} \Delta(\tau)}_{\text{jump costs}} = \|u(s) - v(s)\|_H^2 + \int_s^t \partial_\tau \|u - v\|^2 \, d\tau, \tag{36}$$

where $J(v)$ is the (at most countable) jump set of v, and $\Delta(\tau) = \inf_\gamma \int_0^1 [\|\dot{\gamma}\|^2 + \|u(\tau) - \gamma\|^2] \, dr$ is the optimal transition cost through $\overline{\mathcal{A}}$ at jump time τ.

The key advantage over "ε-division" is that (36) is obtained by *lower semicontinuity* of the energy-dissipation functional as $\varepsilon \to 0$ (chain rule + liminf), not by algebraic manipulation. This is precisely the mechanism established in [32] for vanishing-viscosity limits of gradient systems.

Remark 5.30 (Recovery via Young-measure relaxation)*.* The Γ-convergence of $\mathcal{F}_\varepsilon$ requires recovery sequences $a_\varepsilon \in \mathcal{A}_T$ with $a_\varepsilon \to a$ and $\mathcal{F}_\varepsilon[a_\varepsilon] \to \mathcal{F}_0[a]$. Since $\mathcal{A}_T$ is non-convex and possibly not path-connected, the standard convex-combination construction fails. Two viable alternatives exist:

1. **Young-measure relaxation:** Allow microscopically "chattering" selections (rapid oscillations between attractor points) and work on the relaxed hull $\mathcal{Y}(\mathcal{A}_T)$ of measure-valued trajectories. The Γ-limit exists on $\mathcal{Y}(\mathcal{A}_T)$, and a subsequent *selection theorem* (e.g. Artstein's theorem [39]) extracts a single-valued limit under additional regularity.

2. **Penalise-then-project:** Construct recovery sequences in $L^2(0,T;H)$ (no constraint), then project onto $\mathcal{A}_T$ via Moreau–Yosida regularisation. This requires *prox-regularity* of $\mathcal{A}_T$ in the sense of Poliquin–Rockafellar–Thibault [40], which is weaker than positive reach but stronger than mere closedness.

For the NS trajectory attractor, approach (1) is more natural, since trajectory attractors carry a natural probability structure (statistical solutions / Vishik–Fursikov measures) that provides the required Young-measure framework.

Remark 5.31 (Trajectory attractors vs. classical attractors). For 3D Navier–Stokes, the classical compact attractor $A \subset H$ (Temam) requires uniqueness of solutions, which is open. Trajectory attractors in the sense of Chepyzhov–Vishik [37] and Vishik–Fursikov statistical solutions provide a compact invariant set of *trajectories* without uniqueness—precisely the object needed for $\mathcal{A}_T$. This is a weaker but available starting point.

Remark 5.32 (Independence from geometric regularity). The bound uses only:

- compactness of the trajectory attractor in $L^2_{\mathrm{loc}}(0,T;H) \cap L^2(0,T;V)$,
- the Navier–Stokes energy inequality (for spatial regularity),
- Aubin–Lions compactness (for the $\varepsilon \to 0$ limit).

No positive reach of A, no spectral gap condition, and no inertial manifold is required. This completely circumvents the No-Go zone.

Remark 5.33 (DS3 as the remaining obstruction). In the language of the Dissipative Selection Principle [21], Step 2 establishes DS1–DS2, while Step 3 requires DS3 (precompactness). For 3D Navier–Stokes, DS3 reduces to the Aubin–Lions bound (35), which holds *if* trajectory-attractor elements have uniform H^1-regularity. This is a much more concrete condition than the original Assumption G.

Remark 5.34 (AGC2 strengthened to H^1: a third conditional route via Serrin). Suppose the attractor satisfies $M_{\mathcal{A}} := \sup_{v\in\mathcal{A}} \|v\|_{H^1} < \infty$, and that the strengthened exponential convergence

$$\mathrm{dist}_{H^1}(u(t), \mathcal{A}) \le Ce^{-\lambda t} \quad (t \ge t_0)$$

holds for some $C, \lambda > 0$ (AGC2 in the H^1-metric). Then $\|u(t)\|_{H^1} \le M_{\mathcal{A}} + Ce^{-\lambda t}$, so $u \in L^\infty(t_0, \infty; H^1(\mathbb{T}^3))$. The Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$ then gives $u \in L^\infty_t L^6_x$, which satisfies the Prodi–Serrin criterion $(r, q) = (\infty, 6)$: $\frac{2}{\infty} + \frac{3}{6} = \frac{1}{2} \le 1$. Global regularity follows from Serrin's theorem, without invoking AGC1 or an inertial manifold.

Caveat. Standard Temam attractor convergence (AGC2) holds in the L^2-metric. Strengthening it to H^1 is, for general Leray–Hopf solutions, essentially equivalent to assuming eventual H^1-regularity—the very conclusion to be proved, hence circular. The route is non-circular for *damped or hyperviscous* variants ($\beta > 3$), where H^1-tracking of the attractor can be established independently.

Remark 5.35 (Connection to Kevrekidis–Titi). Diffusion maps [33] approximate numerically the gradient flow of $\mathcal{F}_\varepsilon$ on a data-driven manifold. The rigorous step is to identify the diffusion operator as a discrete approximation of the EDP flow; then BV regularity is the discrete version of inequality (34).

Remark 5.36 (Cross-problem structure). This EDP mechanism is structurally identical to chameleon screening as Γ-convergent phase separation (Dark Energy, [34]) and RG-flow contraction via subadditive ergodic theory (Yang–Mills, [35]). All share the principle: *local control + dissipation ⇒ global selection in the limit.*

Remark 5.37 (Numerical validation on ODE attractors: scope and limitations). **Scope.** The following numerical tests are performed on *finite-dimensional ODE systems* (Lorenz, Rössler, Chen), which have smooth, finite-dimensional attractors, global Lipschitz nonlinearities, and no turbulent cascade. These tests serve as a **proof-of-concept** for the formal algorithm (rescaled EDP functional, BV selection, balanced viscosity), but they do *not* test the mathematical core difficulties of the infinite-dimensional NS setting: Aubin–Lions–Simon compactness in infinite dimension, prox-regularity of infinite-dimensional attractor projections, or log-Lipschitz control of nearest-point maps. A more informative test would use Galerkin-truncated 3D Navier–Stokes simulations (e.g., $N = 32$–64 modes); this is left for future work.

Results. The rescaled functional $\mathcal{F}_\varepsilon$ was numerically minimised on the Lorenz attractor ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) for $\varepsilon \in \{10, 1, 0.1, 0.01, 0.001, 0.0001\}$. The total variation $\mathrm{TV}(v_\varepsilon)$ saturates at ≈ 3682 (compared to $\mathrm{TV}(u) \approx 3662$ for the reference trajectory), with less than 0.3% growth from $\varepsilon = 0.01$ to $\varepsilon = 0.0001$. This confirms DS3 (precompactness) numerically for the ODE setting: the sublevel sets $\{\mathrm{TV} \leq C\}$ are uniformly bounded, and the minimiser v_ε converges to a BV selection on the attractor. Numerical scripts: `compute_ds3_lorenz.py`.

Remark 5.38 (Balanced viscosity on the Lorenz attractor). The balanced-viscosity selection v_ε from Definition 5.29 was tested on the Lorenz attractor ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) using a time-averaged Moreau–Yosida projection. Five tests confirm the theoretical predictions: (i) Simon BV compactness: $\mathrm{TV}(v_\varepsilon)$ remains bounded as $\varepsilon \to 0$ (variation $< 7\%$ for window $W = 50$); (ii) Balanced viscosity convergence: successive differences $\|v_{\varepsilon_1} - v_{\varepsilon_2}\|_{L^2}$ form a Cauchy sequence $(0.171, 0.171, 0.034, 0.004)$; (iii) Aubin–Lions regularity: $\|v_\varepsilon\|_{L^2}$ and $\|v_\varepsilon'\|_{L^2}$ bounded; (iv) Condition D: dissipation integral finite ($\approx 7 \times 10^5$). Script: `compute_bv_selection.py`.

Remark 5.39 (Multi-attractor validation). The balanced-viscosity selection was tested on three chaotic attractors: Lorenz ($\sigma = 10$, $\rho = 28$), Rössler ($a = 0.2$, $c = 5.7$), and Chen ($a = 35$, $c = 28$). All three pass the Simon BV boundedness test ($\mathrm{TV}(v_\varepsilon)$ bounded as $\varepsilon \to 0$), exhibit Cauchy convergence in L^2 (geometric decay of successive differences), and have finite dissipation integrals (Condition D). The Rössler attractor shows the fastest convergence (differences $0.057 \to 0.001$), while the Chen attractor has the highest dissipation ($\approx 1.4 \times 10^6$) but converges nonetheless. This confirms that the BV selection mechanism is robust across different attractor topologies. Script: `compute_bv_multi_attractor.py`.

6 Towards an H^1-Level Obstruction

The dissipation gap identified in Remark 3.36—the finiteness of $\int_0^{T^*} \|\nabla u\|_{L^2}^2\, dt$ even under blow-up—is a fundamental barrier at the L^2-level. In this section we outline a strategy that circumvents this barrier by lifting the attractor-distance functional from L^2 to H^1. The key observation is that the Leray energy identity controls the *energy dissipation* $\int \|\nabla u\|^2$ but *not* the *enstrophy dissipation* $\int \|\Delta u\|^2$; the latter has no a priori upper bound under blow-up and can therefore serve as the divergent dissipative term that drives the contradiction.

6.1 The H^1-distance functional

Definition 6.1 (H^1-attractor distance)**.** For $u \in H^1(\Omega)$, define

$$H^{(1)}(u \mid \mathcal{A}) := \inf_{v \in \mathcal{A}} \frac{1}{2} \|\nabla(u - v)\|_{L^2}^2. \tag{37}$$

Since $\mathcal{A}$ is compact in $H^1(\Omega)$ (Proposition 2.2(i)), the infimum is attained. We write $v^{(1)}(t) \in \mathcal{A}$ for a measurable selection minimising (37) at time t, and $w = u - v^{(1)}$.

Remark 6.2 (Existence of the H^1-minimiser)*.* The minimisation in (37) is over $\mathcal{A}$ with respect to the H^1-seminorm. Since $\mathcal{A}$ is compact in H^1 (Proposition 2.2(i))—note: *compactness*, not convexity, is the operative property—the infimum is attained for each $u \in H^1$. A measurable selection $t \mapsto v^{(1)}(t)$ exists by the Kuratowski–Ryll-Nardzewski theorem applied to the compact-valued multifunction $t \mapsto \arg\min_{v\in\mathcal{A}} \|\nabla(u(t) - v)\|_{L^2}$; this gives L^2-measurability of the selection a priori, with H^1-measurability following from the continuous embedding $H^1 \hookrightarrow L^2$.

6.2 The H^1-energy inequality

Proposition 6.3 (H^1-Gronwall inequality)**.** *Let $u(t)$ be a strong solution of the Navier–Stokes equations on $[0, T)$ and let $v^{(1)}(t) \in \mathcal{A}$ be an H^1-minimising selection. Write $w = u - v^{(1)}$ and $H^{(1)}(t) = \frac{1}{2}\|\nabla w(t)\|_{L^2}^2$. Then*

$$\frac{d}{dt}H^{(1)}(t) \;\le\; -\frac{\nu}{2}\|\Delta u(t)\|_{L^2}^2 \;+\; C_1^{(1)}\, H^{(1)}(t) \;+\; C_2^{(1)} \;+\; \text{minimiser correction}, \tag{38}$$

where $C_1^{(1)} = C(\|\nabla v^{(1)}\|_{L^\infty}, \|\nabla^2 v^{(1)}\|_{L^2})$ and $C_2^{(1)} = C(\|v^{(1)}\|_{H^2}, \|f\|_{H^1})$ are uniformly bounded by Proposition 2.2 (bootstrap regularity gives $\sup_{v\in\mathcal{A}} \|v\|_{H^k} < \infty$ for every k).

Proof sketch. Compute $\frac{d}{dt}\frac{1}{2}\|\nabla w\|^2 = \langle \nabla w, \nabla(\partial_t u - \partial_t v^{(1)})\rangle$. The viscous term yields, via integration by parts ($\langle \nabla w, \nabla \Delta u\rangle = -\langle \Delta w, \Delta u\rangle = -\|\Delta u\|^2 + \langle \Delta v^{(1)}, \Delta u\rangle$):

$$\nu\langle \nabla w,\, \nabla \Delta u\rangle \;=\; -\nu\|\Delta u\|_{L^2}^2 + \nu\langle \Delta v^{(1)},\, \Delta u\rangle.$$

By Young's inequality, the cross-term satisfies $|\nu\langle \Delta v^{(1)}, \Delta u\rangle| \le \frac{\nu}{4}\|\Delta u\|^2 + \nu\|\Delta v^{(1)}\|^2$, giving a net dissipation of $-\frac{3\nu}{4}\|\Delta u\|^2$ plus the bounded attractor-regularity term $\nu\|\Delta v^{(1)}\|^2 \le \nu \sup_{v\in\mathcal{A}} \|v\|_{H^2}^2$.

The nonlinear term $\langle \nabla w,\, \nabla[(u\cdot\nabla)u]\rangle$ is bounded using $\|(u\cdot\nabla)u\|_{H^1} \le C\,\|u\|_{H^1}\,\|u\|_{H^2}$ (product rule and Sobolev embedding $H^2(\mathbb{T}^3) \hookrightarrow L^\infty(\mathbb{T}^3)$; see [4]), giving $|\langle \nabla w, \nabla[(u \cdot \nabla)u]\rangle| \le C\,\|u\|_{H^1}\,\|u\|_{H^2}\,\|\nabla w\|$. Since $\|u\|_{H^2}^2 \le C(\|\Delta u\|^2 + \|u\|^2)$, Young's inequality with parameter $\varepsilon = \nu/4$ yields $C\,\|u\|_{H^1}\,\|\Delta u\|\,\|\nabla w\| \le \frac{\nu}{4}\|\Delta u\|^2 + C_\varepsilon\,\|u\|_{H^1}^2\,\|\nabla w\|^2$. After absorbing this $\frac{\nu}{4}\|\Delta u\|^2$ into the remaining $\frac{3\nu}{4}\|\Delta u\|^2$ budget, the net dissipation is $-\frac{\nu}{2}\|\Delta u\|^2$ (matching the theorem statement), at the cost of a Gronwall factor $C_\varepsilon\,\|u\|_{H^1}^2 = C_\varepsilon(2H^{(1)} + \|v^{(1)}\|_{H^1}^2)$ multiplying $H^{(1)}$. This yields a differential inequality of *Bihari type* (nonlinear Gronwall), since the coefficient $C_1^{(1)}$ depends on $H^{(1)}$ itself:

$$\frac{d}{dt}H^{(1)} \;\le\; -\frac{\nu}{2}\|\Delta u\|^2 + \alpha\, H^{(1)} + \beta\,(H^{(1)})^2 + C_2^{(1)} + R^{(1)},$$

where α, β depend on attractor norms only. The quadratic term $(H^{(1)})^2$ arises because $C_1^{(1)} \sim \|u\|_{H^1}^2 \sim H^{(1)} + \text{const}$. The standard Gronwall lemma does *not* apply directly; instead, Bihari's inequality gives a finite-time bound on $H^{(1)}$ provided the dissipation term $\frac{\nu}{2}\|\Delta u\|^2$ controls the quadratic growth. **Open point:** Under blow-up, whether the quadratic Gronwall cost $\beta(H^{(1)})^2$ grows slower than the divergent enstrophy dissipation $\int \|\Delta u\|^2$ is itself part of the open question; the Bihari bound only provides control when the ODE does *not* reach its explosion time before the PDE does. A complete argument requires either (a) a quantitative blow-up rate estimate showing $\int \|\Delta u\|^2$ diverges faster than any polynomial in $(T^* - T)^{-1}$, or (b) an independent upper bound on $H^{(1)}$ near T^* that prevents the Bihari cost from exploding. The differential inequality (40) in the theorem statement should therefore be understood as valid on every compact subinterval $[0, T'] \subset [0, T^*)$ where $\|u\|_{H^1}$ remains finite (i.e., in the strong-solution regime), which is precisely the regime relevant to the proof by contradiction.

The minimiser correction term is $R^{(1)}(t) := |\langle \nabla w, \nabla \partial_t v^{(1)} \rangle| \leq \|\nabla w\|_{L^2} \|\nabla \partial_t v^{(1)}\|_{L^2} \leq \sqrt{2H^{(1)}}\, L^{(1)}(t)$, where $L^{(1)}(t) := \|\partial_t v^{(1)}(t)\|_{H^1}$ is the temporal regularity bound from Assumption $G^{(1)}$. Under $G^{(1)}$, $L^{(1)} \in L^1_{\mathrm{loc}}$, so $\int_0^T R^{(1)}\, dt \leq \sqrt{2 \sup H^{(1)}} \int_0^T L^{(1)}\, dt < \infty$ on every compact subinterval of $[0, T^*)$. □

6.3 Assumption $G^{(1)}$ and the enstrophy dissipation

Definition 6.4 (Assumption $G^{(1)}$)**.** Let $u(t)$ be a strong solution on $[0, T)$ with global attractor $\mathcal{A} \subset H^1$. For a.e. $t \in [0, T)$, let $v^{(1)}(t) \in \arg\min_{v \in \mathcal{A}} \|\nabla(u(t) - v)\|_{L^2}^2$. We say that **Assumption** $G^{(1)}$ holds if:

($G^{(1)}$.1) **Uniform attractor regularity.** $\sup_{v \in \mathcal{A}} \|v\|_{H^2} < \infty$. (This follows from the known smoothness of the attractor; cf. Proposition 2.2.)

($G^{(1)}$.2) **Temporal regularity of the projection.** The map $t \mapsto v^{(1)}(t)$ has bounded variation in H^1 on every compact subinterval $[0, T'] \subset [0, T)$, and there exists $L^{(1)} \in L^1_{\mathrm{loc}}(0, T)$ such that

$$\|\partial_t v^{(1)}(t)\|_{H^1} \;\leq\; L^{(1)}(t) \qquad \text{for a.e. } t.$$

This assumption is the exact H^1-analogue of Assumption G in the L^2-case; it concerns exclusively the temporal stability of the attractor projection and imposes no additional regularity on the Navier–Stokes solution itself.

Remark 6.5 (Why $G^{(1)}$ is structurally the same as G). Assumption $G^{(1)}$ isolates the same geometric difficulty as Assumption G, but at the correct Sobolev level. It has exactly the same form as G (Definition 3.5), lifted from L^2 to H^1. The attractor regularity (Proposition 2.2) guarantees $\sup_{v \in \mathcal{A}} \|v\|_{H^k} < \infty$ for all k, so the *spatial* regularity of $v^{(1)}$ is not an issue. The difficulty, as before, is exclusively the *time* regularity: whether the H^1-minimiser jumps or evolves smoothly. The AGC and TLL/LDI frameworks (Section 3) transfer directly, with the Lipschitz/BV conditions reformulated in H^1. $G^{(1)}$ is stronger than G (it requires control of one additional derivative), but not qualitatively different.

6.4 Why the H^1-level avoids the dissipation gap

The crucial asymmetry is:

Level	Dissipation integral	Under blow-up
L^2 (energy)	$\int_0^{T^*} \|\nabla u\|_{L^2}^2\, dt$	**Finite** (Leray identity)
H^1 (enstrophy)	$\int_0^{T^*} \|\Delta u\|_{L^2}^2\, dt$	**No a priori bound**

This is not a technical detail but a *fundamental asymmetry* in the Navier–Stokes energy hierarchy: viscous dissipation at the energy level ($\int \|\nabla u\|^2$, physically: enstrophy production) is controlled by the initial energy budget, while viscous dissipation at the enstrophy level ($\int \|\Delta u\|^2$, physically: palinstrophy production) is *not*. This asymmetry is the structural reason why the L^2-level argument cannot produce a contradiction while the H^1-level argument can.

The H^1-energy identity reads

$$\frac{1}{2}\|\nabla u(T)\|^2 + \nu \int_0^T \|\Delta u\|^2 \, dt \;=\; \frac{1}{2}\|\nabla u_0\|^2 + \int_0^T \Big[\langle (u \cdot \nabla)u,\, \Delta u\rangle + \langle \nabla f,\, \nabla u\rangle\Big] \, dt. \quad (39)$$

Under blow-up, $\|\nabla u(T)\|^2 \to \infty$ as $T \nearrow T^*$. The left-hand side therefore diverges. Since the nonlinear term $\int \langle (u \cdot \nabla)u, \Delta u\rangle$ can grow without bound (it involves $\|u\|_{H^1} \|\Delta u\|$), the balance between the three terms on the right is *not* controlled a priori. In particular, $\int_0^{T^*} \|\Delta u\|^2 \, dt$ may diverge—and if it does, the $H^{(1)}$-functional is driven below zero by exactly the same budget argument as in Theorem 3.35, but now with a *genuinely divergent* dissipation term.

6.5 Conditional H^1-theorem (sketch)

Proposition 6.6 (Conditional H^1-obstruction to finite-time blow-up (sketch))**.** ***Status:*** *This is a* proof sketch*, not a complete proof. The three gaps identified in Remark 6.7 below (projection regularity, enstrophy-dissipation divergence, Bihari control) are each individually open problems of comparable difficulty to the Navier–Stokes regularity problem itself. The result is presented to motivate the H^1-level programme, not as an established theorem.*

Let $u(t)$ be a strong solution of the 3D Navier–Stokes equations on $[0, T^*)$ with smooth initial data and smooth forcing. Assume that Assumption $G^{(1)}$ (Definition 6.4) holds. Define the H^1-distance functional $H^{(1)}(u(t) \mid \mathcal{A}) := \inf_{v \in \mathcal{A}} \frac{1}{2}\|\nabla(u(t) - v)\|_{L^2}^2$.

Then along the solution, the differential inequality

$$\frac{d}{dt} H^{(1)}(u(t) \mid \mathcal{A}) \;\le\; -\frac{\nu}{2}\|\Delta u(t)\|_{L^2}^2 + C_1(t)\, H^{(1)}(u(t) \mid \mathcal{A}) + C_2 + R^{(1)}(t) \quad (40)$$

holds, where $C_1(t) = C(\|u(t)\|_{H^1}^2) = C(2H^{(1)}(t) + \|v^{(1)}\|_{H^1}^2)$ depends on the solution norm (making the inequality of Bihari type), $C_2 < \infty$ depends only on attractor norms, and the correction term $R^{(1)}$ is locally integrable under Assumption $G^{(1)}$.

If additionally

$$\int_0^{T^*} \|\Delta u(t)\|_{L^2}^2 \, dt = \infty, \quad (41)$$

then $H^{(1)}(u(t) \mid \mathcal{A})$ cannot remain non-negative for all $t < T^*$. In particular, a finite-time blow-up at $T^* < \infty$ is excluded under these assumptions.

Proof sketch. Integrate (40) on $[0, T]$ for $T < T^*$:

$$H^{(1)}(T) \;\le\; H^{(1)}(0) - \frac{\nu}{2}\int_0^T \|\Delta u\|^2 \, dt + \int_0^T \Big(C_1 \, H^{(1)} + C_2\Big) \, dt + \int_0^T R^{(1)} \, dt.$$

Under $G^{(1)}$, the Gronwall cost and the correction $\int R^{(1)}$ are finite on every compact subinterval of $[0, T^*)$. Note that the Gronwall coefficient C_1 depends on $\|u\|_{H^1}^2 \sim H^{(1)} + \text{const}$, making the inequality of Bihari type (nonlinear Gronwall). In the proof-by-contradiction regime (u is a strong solution on $[0, T^*)$), $\|u\|_{H^1}$ is finite for every $T < T^*$, so the integrated cost $\int_0^T (C_1 H^{(1)} + C_2) \, dt$ is finite for each fixed $T < T^*$ (though it may grow as $T \nearrow T^*$).

By (41), the dissipation term $\frac{\nu}{2}\int_0^T \|\Delta u\|^2\,dt \to \infty$ as $T \nearrow T^*$. The conclusion requires that the divergent dissipation outpaces the (growing) Gronwall cost. Because the Gronwall coefficient $C_1^{(1)} \sim H^{(1)}$ makes this a *Bihari-type* inequality, the Gronwall cost itself may grow super-exponentially near T^*. Whether the enstrophy dissipation diverges fast enough to dominate this growth is an open quantitative question (see Remark 6.7, gap 3). If it does, then there exists $T_0 < T^*$ such that $H^{(1)}(T_0) < 0$, contradicting $H^{(1)} \geq 0$.

Caveat. This argument is not a complete proof. In particular: (a) the hypothesis (41) is itself an open problem (see Remark 6.7, gap 2); (b) the Bihari structure means the Gronwall cost may grow as fast as the dissipation, so the domination is not automatic; and (c) Assumption $G^{(1)}$ is open. □

Remark 6.7 (Status and comparison with L^2-theorem). Proposition 6.6 replaces the dissipation-dominance condition (D) of Theorem 3.35 with the more concrete hypothesis (41) (divergent enstrophy dissipation). The conditional structure now has *three* gaps:

1. **Projection gap:** Assumption $G^{(1)}$ (regularity of the H^1-minimiser)—same character as Assumption G, same attack vectors (AGC, TLL/LDI).

2. **Enstrophy-dissipation gap:** Whether (41) holds under BKM blow-up. Unlike the L^2-case, there is *no* a priori bound ruling this out. Establishing (41) requires understanding the nonlinear transfer at the enstrophy level (vortex stretching vs. viscous damping).

3. **Bihari structure:** Unlike the L^2-theorem (where C_1 is a constant depending only on attractor norms), the H^1-Gronwall coefficient $C_1^{(1)}$ depends on $\|u\|_{H^1}^2 \sim H^{(1)}$, yielding a Bihari-type (nonlinear Gronwall) inequality. In the proof-by-contradiction setting, $H^{(1)}$ is finite for each $T < T^*$ (strong solution), so the integrated Gronwall cost is finite; but the growth rate of this cost near T^* must be slow enough relative to the diverging dissipation (41). This is a quantitative refinement of gap (2) and is automatically satisfied if $\int \|\Delta u\|^2$ diverges faster than any polynomial.

At L^2-level, the analogous contradiction is impossible because the Leray energy identity forces $\int_0^{T^*} \|\nabla u\|_{L^2}^2 < \infty$ always. At H^1-level, no corresponding a priori bound for $\int \|\Delta u\|_{L^2}^2$ exists; this structural asymmetry is what makes the H^1-lift viable in principle.

Proposition 6.6 is doubly conditional: it isolates two independent open questions—the temporal regularity of the H^1-projection ($G^{(1)}$) and the divergence of the enstrophy dissipation under blow-up—without presupposing either.

Open problems for closing the H^1-level gaps:

1. Prove $G^{(1)}$ from attractor geometry (parallel to the L^2 AGC programme).

2. Prove (41) from the enstrophy balance and BKM blow-up. A sufficient condition would be: the vortex stretching term $\int_\Omega (\omega\cdot\nabla)u\cdot\omega\,dx$ grows at most like $C\,\|\Delta u\|^{2-\delta}$ for some $\delta > 0$ near T^*.

3. Alternatively: lift the TLL/LDI framework to H^1, yielding BV selections for the H^1-projection. This would give a BV variant of $G^{(1)}$ without inertial manifolds.

6.6 Post-Zookeeper transfer: damped Navier–Stokes and bad mechanisms

The post-Zookeeper role of this framework is not to remove the two conditional assumptions at once. Instead, it gives a sharper route for what must be proved next. The Navier–Stokes transfer slots are:

This is an RFEP v1.8 normal-form transfer, not an automatic PDE regularity theorem. The Zookeeper closure removes the Riemann $C^{(2)}$/MS2 blocker inside CORE; the Navier–Stokes branch still has to verify the operator/form, defect, Galerkin, gap, and limit obligations in its own analytic setting.

Operator/form. The Stokes semigroup, Leray projection, and nearest-attractor selection map.

Defect. The attractor-distance budget $H(u \mid \mathcal{A})$, the projection-jump defect, the dissipation defect D, and a separate high-frequency injection defect.

Approximation. Fourier–Galerkin truncations, damped Navier–Stokes systems, hyperviscous systems, and finite determining modes.

Gap. Squeezing/TLL/LDI constants, damping gaps, attractor reach, and enstrophy barriers.

Limit. Passage from damped or hyperviscous test systems to the classical $\beta = 1$ three-dimensional equation.

The constructive intermediate target is a damped 3D Navier–Stokes theorem. In regimes where damping gives better attractor regularity and uniqueness, one should prove

$$\text{damping gap} + \text{attractor regularity} \Longrightarrow \text{TLL/LDI} \Longrightarrow \text{BV selection}.$$

This would not solve the Millennium problem, but it would turn the current proof-of-life evidence into a genuine PDE-scale transfer theorem.

The negative side should be equally explicit. Recent blow-up and non-uniqueness scenarios motivate a "bad-mechanism catalogue": Type-I profiles, instantaneous blow-up, inverse cascades, high-to-low frequency injection, and weak–strong failure should each be tested against H, D, and the projection-variation budget. This makes the conditional framework falsifiable: a credible regularity route must say which of these mechanisms is excluded, by which defect, and at which scale.

Proof status summary

This paper is a **doubly conditional framework**: it shows that global regularity *follows from* two open conditions—Assumption G (attractor-projection regularity) and Condition D (dissipation dominance)—without claiming that either condition is easier to verify than regularity itself. The value lies in the decomposition, not in a simplification.

Component	Status
Proven (unconditional):	
$H(u\|\mathcal{A})$ Gronwall inequality	Lemma (rigorous)
Assumption G + Condition D $\Rightarrow$ regularity	Theorem (rigorous)
G' (BV) suffices for G	Proposition (rigorous)
TLL + LDI $\Rightarrow$ G' (BV chain rule)	Lemma (rigorous)
Conjectural / proof sketch:	
Squeezing $\Rightarrow$ TLL	Conjecture (open step: log-Lip bound)
H^1-obstruction (Prop. 6.6)	Proof sketch (3 open gaps)
Numerically confirmed (ODE systems only):	
BV selection on Lorenz/Rössler/Chen	3/3 passed (finite-dim.)
Balanced viscosity convergence	Cauchy sequence (finite-dim.)
Condition D (dissipation bounded)	3/3 passed (finite-dim.)
Open (the core problems):	
Selection Regularity (Assumption G)	Open
Condition D at L^2-level	Structurally limited (see Remark 3.36)
Enstrophy-dissipation divergence (H^1)	Open
Determining modes $\Rightarrow$ squeezing	Open

7 Conclusion

By formulating the regularity problem as a distance-dissipation balance relative to the global attractor of the Navier–Stokes semiflow, we establish a *doubly conditional* framework for regularity. This framework does not claim to reduce the regularity problem to simpler components; rather, it decomposes the question into two structurally distinct open problems, each admitting independent investigation. The two conditions are:

1. **Assumption G** (or the weaker G$'$): the time-dependent attractor projection has sufficient regularity (AC or BV).

2. **Condition (D)**: the cumulative viscous dissipation dominates the Gronwall cost near a hypothetical blow-up time (Remark 3.36).

Under both conditions, finite-time blow-up leads to $H(u \mid \mathcal{A}) < 0$—a contradiction to its non-negativity.

The dissipation gap. The Leray energy identity guarantees that $\int_0^{T^*} \|\nabla u\|_{L^2}^2 \, dt < \infty$ even under blow-up (Remark 3.4). What diverges is the *pointwise* enstrophy $\|\nabla u(t)\|_{L^2}^2 \to \infty$, not its time-integral. Whether this pointwise divergence is fast enough to overcome the Gronwall cost is the content of condition (D)—a structural question about blow-up rates that is open for the 3D Navier–Stokes equations.

Two parallel attack paths for Assumption G are identified: (i) the AGC path, reducing G to positive reach of the attractor (equivalent to inertial manifold existence); (ii) the log-Lipschitz path (Lemma 3.20), reducing G$'$ to trajectory-level regularity (TLL) combined with log-distance integrability (LDI). The second path avoids the inertial manifold barrier entirely.

Three strategies for condition (D) are outlined in Remark 3.36: lifting the functional to H^1-level, exploiting quantitative blow-up rates, or using Serrin-type criteria. Closing either the projection gap (Assumption G) *or* the dissipation gap (condition D) would advance the present framework, and closing both would yield an unconditional regularity proof.

The H^1-level programme (Section 6) offers the most promising path forward: by lifting the attractor-distance functional from L^2 to H^1, the dissipation term becomes $\int \|\Delta u\|^2$, which—unlike $\int \|\nabla u\|^2$—has *no* a priori upper bound under blow-up. This structural asymmetry between energy and enstrophy dissipation is the key insight of the present work and motivates Proposition 6.6 as the natural continuation of this programme. At the H^1-level, a further subtlety emerges: the Gronwall coefficient $C_1^{(1)}$ depends on $\|u\|_{H^1}^2$, yielding a Bihari-type (nonlinear Gronwall) differential inequality whose integrated cost grows as the solution approaches the hypothetical blow-up. The proof by contradiction remains valid provided the divergent enstrophy dissipation outpaces this growth—a quantitative refinement that connects the H^1 programme to precise blow-up rate estimates.

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